

A resource-stress model resolves context-dependent chromosome instability and ploidy evolution across *in vitro* and *in vivo* cancer models

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Abstract

Whole-genome doubling and aneuploidy are common in cancer, but the fitness of high-ploidy states depends on biological context. We analyzed matched near-diploid and near-tetraploid SUM159 lineages with a chromosome-number-structured model calibrated to passage growth, chromosome-number and flow-density data *in vitro*, and to longitudinal tumor burden, terminal chromosome-number profiles, and terminal necrosis *in vivo*. The separate *in vitro* fit reproduced ploidy reshaping during oxygen deprivation without requiring substantial direct hypoxia-origin death. Across 500 separate *in vivo* fits, fixed-O₂ attractor analyses retained multiple compatible oxygen–ploidy response regimes. For joint fitting, six landscape-informed *in vivo* warm starts were each paired with the same best separate *in vitro* fit and optimized from 500 numerical seeds. Pair-balanced summaries consistently estimated a lower maximal division rate and lower oxygen-response threshold *in vivo*, together with higher fitted stress-associated death and chromosome-missegregation parameters. In contrast, the three post-missegregation survival parameters did not support one uniform functional shift across all warm-start pairs. These contrasts are conditional on the fitted model and quantify numerical stability across optimization starts; they are not biological replicates or direct causal measurements of oxygen. The results show that a common mechanistic model can reconcile culture and tumor observations while retaining alternative parameter regimes that must be resolved by additional data.

Introduction

Whole-genome doubling (WGD) and aneuploidy are common features of cancer evolution, yet their prevalence varies substantially across biological contexts.^{1–3} A central unresolved question is why high-ploidy states are tolerated in some environments but selected against in others.^{4,5} One clue comes from the striking difference between cultured cancer cell lines and primary tumors: across organ systems, cell lines exhibit markedly higher ploidy than primary tumors. This observation suggests that standard *in vitro* culture conditions, which provide abundant nutrients and simplified

environmental constraints, may relax selection against high-ploidy states, whereas in vivo tumor environments impose stronger constraints on cells carrying excess genomic material.^{4,5}

A plausible basis for this context dependence is the energetic cost of increased chromosome content.⁴⁻⁶ WGD increases the biosynthetic burden associated with DNA replication, RNA production, and especially protein synthesis.⁴⁻⁶ Aneuploidy further compounds this cost by disrupting gene dosage balance, producing proteotoxic, osmotic, oxidative, replication, and endoplasmic-reticulum stress.⁷⁻⁹ These stresses require active homeostatic responses, including protein quality control, degradation of excess proteins, DNA repair, and metabolic adaptation.^{5,7-9} Thus, high-ploidy cells are expected to carry a resource-dependent fitness cost, particularly when oxygen, glucose, or other metabolic substrates become limiting.^{4,5}

At the same time, resource scarcity can also increase genomic instability.^{4,5} Environmental stresses such as serum starvation, hypoxia, folate deficiency, and energy restriction have been linked to karyotypic instability, chromosome mis-segregation, polyploidization, and increased cellular heterogeneity. The evolutionary consequences of these events depend strongly on ploidy. Chromosome mis-segregation can trigger p53-mediated arrest, senescence, or death, particularly when chromosome loss creates severe dosage imbalance.^{8,10-13} WGD and high-ploidy states can buffer these effects by reducing the relative dosage impact of individual chromosome gains or losses, allowing some mis-segregations to be converted into heritable karyotype variation rather than lethal events.^{7,14,15} Consistent with this view, ALFA-K, a computational method that infers chromosome-level karyotype fitness from longitudinal single-cell karyotype data, showed that mis-segregations impose stronger fitness costs near diploidy than near tetraploidy.¹⁶ Thus, ploidy is a quantitative determinant of whether CIN generates nonviable progeny or heritable karyotypic diversity.

Oxygen provides a tractable first-order proxy for this resource landscape.⁴⁻⁶ Across tissue sites, WGD prevalence correlates with peripheral oxygen partial pressure, consistent with the idea that well-oxygenated tissues are more permissive for WGD-positive cancers, whereas poorly oxygenated tissues are less permissive.^{5,6} However, oxygen is an imperfect surrogate for total energy availability. Tissue resource states also depend on amino acid and glucose availability, local metabolic consumption, and metabolic specialization. For example, tissues that rely heavily on glycolysis or have unusually high glucose availability may not experience low oxygen as equivalent to global energy scarcity.

Together, these observations identify a central tension in cancer ploidy evolution. Resource scarcity should select against WGD and other high-ploidy states because additional chromosome content increases biosynthetic, proteostatic, and homeostatic demands.⁴⁻⁶ Yet adverse environments can also promote CIN, chromosome mis-segregation, and polyploidization, generating the karyotypic variation on which selection acts.^{4,5} Whether this variation is eliminated or retained depends in part on ploidy: near-diploid cells are more vulnerable to lethal dosage imbalance after chromosome loss, whereas WGD or high-ploidy cells can better buffer individual segregation errors and convert them into heritable karyotype variation.^{7,14-16} Thus, resource limitation may simultaneously oppose and promote WGD, depending on how strongly it suppresses proliferation, induces CIN, and filters the survival of altered progeny. A key question is therefore which environments select for WGD, which select against it, and through which mechanisms. Answering this requires a quantitative framework that can jointly represent resource-dependent proliferation, stress-induced chromo-

some instability, and ploidy-dependent survival of resulting karyotypic states.

Methods

Study design and data streams. We analyzed near-diploid and near-tetraploid SUM159 lineages across two experimental contexts. *In vivo*, orthotopic tumors initiated from the near-diploid and near-tetraploid lineages were followed with longitudinal tumor-burden measurements and harvested for terminal single-cell chromosome-number profiling; where matched terminal histology was available, necrosis was quantified at harvest. The *in vivo* calibration used the untreated subset of eight tumors, four initiated from each lineage. *In vitro*, the corresponding lineage pair was analyzed in passage-structured culture experiments with measured passage growth, karyotype/chromosome-number observations, and G0/G1 flow-density profiles under controlled oxygen conditions. The shared lineage origins placed the two datasets on a common modeling axis, but the experiments were not biological replicates of one another and did not constitute a randomized intervention comparing culture with tumors.

Model state and oxygen context. The model represents viable cells as a distribution over total chromosome-number states. It also tracks non-viable biomass in separate dead compartments, so simulated bulk tumor burden includes both live cells and uncleared dead material. Terminal single-cell chromosome-number profiles sample only the viable population at harvest, whereas tumor burden and necrotic fraction depend on live plus uncleared dead compartments. This state representation lets one simulation be compared simultaneously with longitudinal tumor volume, terminal necrosis, and single-cell chromosome-number distributions, while preserving the model’s focus on karyotype evolution.

Resource context is represented differently in the two settings. *In vivo*, oxygen is modeled as an effective mean-field level that declines with chromosome-weighted tumor demand and relaxes toward a supply-demand target. *In vitro*, each passage segment is simulated at its assigned fixed oxygen level.

Mechanistic modules. The mechanistic core of the model has three modules corresponding to the routes through which resource context could select for or against WGD. First, oxygen/resource stress suppresses proliferation through a chromosome-number-dependent growth penalty, so high-chromosome states pay a larger growth cost when resources are limiting. Stress also induces a chromosome-number-dependent live-to-dead death hazard that feeds the hypoxia-origin dead compartment. Second, stress can increase chromosome missegregation, while WGD is modeled as a division-associated conversion branch that provides a route into high-ploidy states, generating the karyotypic variation on which selection can act. Third, daughters produced after missegregation pass through a ploidy-dependent survival filter: daughters of lower-ploidy mothers are more vulnerable to lethal dosage imbalance, whereas higher-ploidy mothers can better buffer chromosome gains or losses and convert some CIN events into heritable karyotype variation. These modules map directly onto the Introduction’s mechanistic alternatives by asking whether resource-poor environments suppress high-ploidy proliferation, generate CIN and WGD, or determine whether altered progeny die or persist.

Observation models. The observation models link each measured data stream to a specific biological component of this mechanism, so the fitted trajectory is constrained by both growth dynamics and chromosome-number evolution:

Data stream	Biological component constrained
Longitudinal tumor burden	Net growth, death, clearance, and resource-dependent proliferation
Terminal single-cell chromosome-number distributions	Karyotype evolution, WGD, missegregation, and survival of altered progeny
Terminal necrotic fraction	Uncleared dead biomass relative to total tumor burden
<i>In vitro</i> passage growth rates	Proliferation under controlled culture and oxygen conditions
<i>In vitro</i> karyotype/chromosome-number and flow-density profiles	Chromosome-number evolution in culture

Joint fitting. We used the same cellular response model for the separate and joint *in vivo*/*in vitro* fits: chromosome-number transitions, hypoxia/resource-stress effects on proliferation and death, stress-induced missegregation, WGD generation, and post-missegregation survival were represented by the same mechanistic structure. The contexts differed in how oxygen trajectories were supplied and in their observation models. In the reported joint analysis, all 14 cross-context mechanistic parameters were soft-coupled. Each received separate *in vivo* and *in vitro* values reconstructed around a transformed-scale center, and a bounded Welsch penalty regularized their separation. The joint fit therefore tested which parameter contrasts were repeatedly recovered across optimizer starts and landscape-informed warm-start pairs. These contrasts are conditional on the model, data streams, regularization, and shared *in vitro* anchor; they identify model-supported hypotheses rather than direct measurements of environmental causes.

Together, this framework allowed us to ask whether environmental differences between tumors and culture select for or against WGD by altering resource-dependent proliferation, stress-induced chromosome missegregation, WGD generation, or the ploidy-dependent survival of karyotypically altered progeny.

Results

Matched culture and tumor trajectories motivate a resource-stress model of ploidy evolution

The motivating observations differed by experimental setting. In culture, longitudinal chromosome-number measurements showed a downward shift after a transient whole-genome-doubling/high-ploidy expansion, demonstrating that the same lineage could first generate and then reduce high-ploidy states. In tumors, longitudinal burden measurements were paired with chromosome-number and histology/necrosis measurements at harvest; the intervening chromosome-number trajectory was inferred by the model rather than observed directly. The near-diploid and near-tetraploid

SUM159 lineage pairs nevertheless placed the *in vitro* and *in vivo* data on a common interpretive axis (Figure 1A), allowing fitted context contrasts to be compared without treating the two experiments as direct biological replicates.

These observations motivated a model in which resource limitation can both oppose and promote high-ploidy states. In the model overview, resource stress imposes growth and death costs on chromosome-rich states, while also increasing CIN/WGD pressure and generating the variation on which selection acts; high ploidy can then buffer some missegregation events that would be lethal or strongly deleterious near diploidy (Figure 2). Thus, the predicted ploidy outcome depends on the balance among growth suppression, stress-associated death, CIN generation, WGD generation, and post-missegregation survival filtering.

In vitro ploidy evolution under hypoxia occurs without strong high-ploidy elimination

Separate *in vitro* fits showed that oxygen stress can reshape ploidy without requiring strong elimination of high-ploidy cells. The fit jointly tracked proliferation, chromosome-count observations, and live/dead burden components across the 2N and 4N control and oxygen-deprived branches, and recapitulated the transient high-ploidy/WGD phase in the oxygen-deprived 2N lineage before later lower-ploidy reshaping (Figure 3A). Low oxygen slowed proliferation and altered ploidy trajectories, but the fit did not require large net death or strong direct removal of high-ploidy states.

The resulting mechanism is therefore not simply that 4N cells are killed off. Instead, high-ploidy cells carry more chromosomes and therefore create more opportunities for chromosome-loss events during division, while the fitted post-missegregation survival function makes a subset of chromosome-loss daughters from high-ploidy mothers viable (Figure 3B–D). This makes buffering double-edged: high-ploidy cells carry more chromosomes and therefore generate more chromosome-loss opportunities during division, while ploidy-dependent post-missegregation survival allows a subset of those chromosome-loss daughters to remain viable. In this model, high-ploidy expansion can therefore seed viable lower-ploidy descendants and produce a trajectory back toward more stable lower-ploidy states without requiring strong direct elimination of the high-ploidy population. Along the terminal severe-deprivation 4N lineage, the fitted mean chromosome number declined from 84.3 to 80.8 while the model-predicted direct hypoxia-origin dead burden remained at or below 1.72% of total burden; dead-buffer losses from chromosome-number transitions and boundary routing were tracked separately (Figure 3E).

In vivo resource regime determines which ploidy state is favored

For the *in vivo* fits, oxygen should be interpreted as the formal model variable and as a first-order proxy for broader effective resource stress. The fitted effective O₂ trajectories are inferred latent variables, not direct oxygen measurements, and they order simulated tumors along a model-implied resource axis (Figure 4A). This distinction is important because *in vivo* hypoxia is not equivalent to *in vitro* low oxygen in rich medium: tumor stress can also reflect nutrient limitation, perfusion heterogeneity, waste buildup, cell density, stromal constraints, and tissue structure.

To ask whether oxygen alone would push fitted tumors toward predictable long-term chromosome-number compositions, we evaluated each fit at fixed O₂ reference levels and recorded the dominant mean-ploidy outcome from the analytical fixed-O₂ attractor calculation (Methods; Supplementary

Methods). We then scored fitted parameters and derived features by how well each separated lower- from higher-ploidy fixed- O_2 outcomes, using one-feature AUCs as compact descriptive indicators of candidate drivers. At low O_2 , stress-associated death terms were the strongest separators of lower- versus higher-ploidy outcomes (Figure 4B). At intermediate O_2 , the outcome was less reducible to a single feature and was associated with death selectivity, missegregation sensitivity, buffering strength, oxygen thresholding, and baseline missegregation (Figure 4C). At high O_2 , baseline missegregation and buffering-related parameters ranked more prominently (Figure 4D). Because both the labels and candidate features were derived from the fitted model, these AUCs rank separation within the fitted landscape; they are not hypothesis tests, effect-size estimates from biological replicates, or evidence that one parameter is a causal driver.

To ask whether fitted solutions occupied distinct regions of parameter space, we also compared fitted *in vivo* parameter sets after transforming and scaling parameters relative to their prior/bound ranges (Methods; Supplementary Methods). Baseline missegregation probability and oxygen-response shape differed descriptively across three solution clusters, while the pooled t-SNE view showed how the selected *in vivo* regions related to the broader *in vivo/in vitro* optimization landscape (Figure 4E–F). These clusters are exploratory summaries of fitted solutions rather than independently validated biological subtypes.

Tumors and culture impose distinct ploidy-selective pressures

Joint soft-coupled fits were used to ask which fitted mechanisms remained context-specific when the SUM159 lineage pairs were compared in culture and tumors (Figure 5A). Six joint warm starts paired representatives from three *in vivo* parameter-landscape clusters and two subclusters per cluster with the same best separate *in vitro* fit (seed 10); each pair was optimized from 500 numerical seeds. All 14 cross-context mechanistic parameters were represented by context-specific values centered on a common transformed-scale value, with a Welsch penalty discouraging small *in vivo/in vitro* separations. Pair-level summaries first aggregated the 500 optimizer starts within each warm-start pair and then compared directions across the six pairs. Optimizer starts quantify numerical stability and are not biological replicates; moreover, all six pairs share the same *in vitro* anchor.

Several parameter directions were stable across the six warm-start pairs (Figure 5B–D). The pair-median *in vivo/in vitro* ratio for the maximal division rate, $\lambda_{\max}$, ranged from 0.099 to 0.223, supporting a lower fitted proliferative ceiling *in vivo*. The maximal stress-induced missegregation increment, p_{misseg} , was higher *in vivo* in all six pairs, with pair-median ratios from 11.1 to 47.4; the *in vivo* value reached its active upper bound in two pairs, so the largest ratios should not be interpreted as precise effect sizes. The oxygen threshold $O_{2,c}$ and buffering-strength parameter β_{buf} were lower *in vivo* in all six pairs, whereas μ_{hp} , γ_{μ} , $k_{o,\text{mis}}$, and γ_{growth} were higher in all six. Several of these axes were also boundary-limited. These are parameter-level contrasts: because the stress-linked missegregation function depends jointly on p_{misseg} , $k_{o,\text{mis}}$, and the death-hazard parameters, the direction of one parameter alone does not establish the direction of the complete response function.

The post-missegregation survival parameters did not support the earlier interpretation of one uniformly more ploidy-dependent survival filter *in vivo*. Although β_{buf} was lower in all six pairs, the dominant ratio class for $s_{\max}$ was split between lower and approximately coupled values, and n_{exp} was higher in four pairs and lower in two. Thus, the current joint results support robust fitted

contrasts in proliferative capacity and several stress/CIN parameters, while the composite survival-filter difference remains warm-start-regime dependent. The 5th–95th percentile intervals across the 500 numerical seeds were narrow within most pairs, but these intervals describe optimizer-start spread, not biological confidence intervals.

These analyses left multiple parameter regimes that could explain the *in vivo* ploidy trajectories. We characterized this uncertainty with fixed- O_2 dominant-eigenvector response curves: for each candidate fit, the analytical attractor was evaluated across an oxygen grid and grouped by the shape of the predicted O_2 –ploidy relationship (Methods; Supplementary Methods). The 500 separate *in vivo* fits occupied several response classes rather than a single selected class (Figure 6A). We therefore also projected every seed- O_2 attractor into the plane defined by fixed oxygen and the population-average per-chromosome missegregation probability, with dominant mean ploidy shown as the response (Figure 6B). This map summarizes where fitted solutions lie; it is not an intervention that varies oxygen and chromosome instability independently. Finally, response class and fit objective were overlaid on a prior-scaled parameter embedding. Several common response classes overlapped in parameter space and objective value, and no single response class uniquely contained the lowest-objective solutions. This diagnostic therefore retained competing regimes rather than selecting one response shape by assumption (Figure 6C).

Optimization data streams for the shared SUM159 2N/4N system

A 3 days start at the first post-MRCA in vitro fit passage and at in vivo measurement day 0. In vitro oxygen is known by passage; in vivo oxygen is not directly measured and is latent in the model.

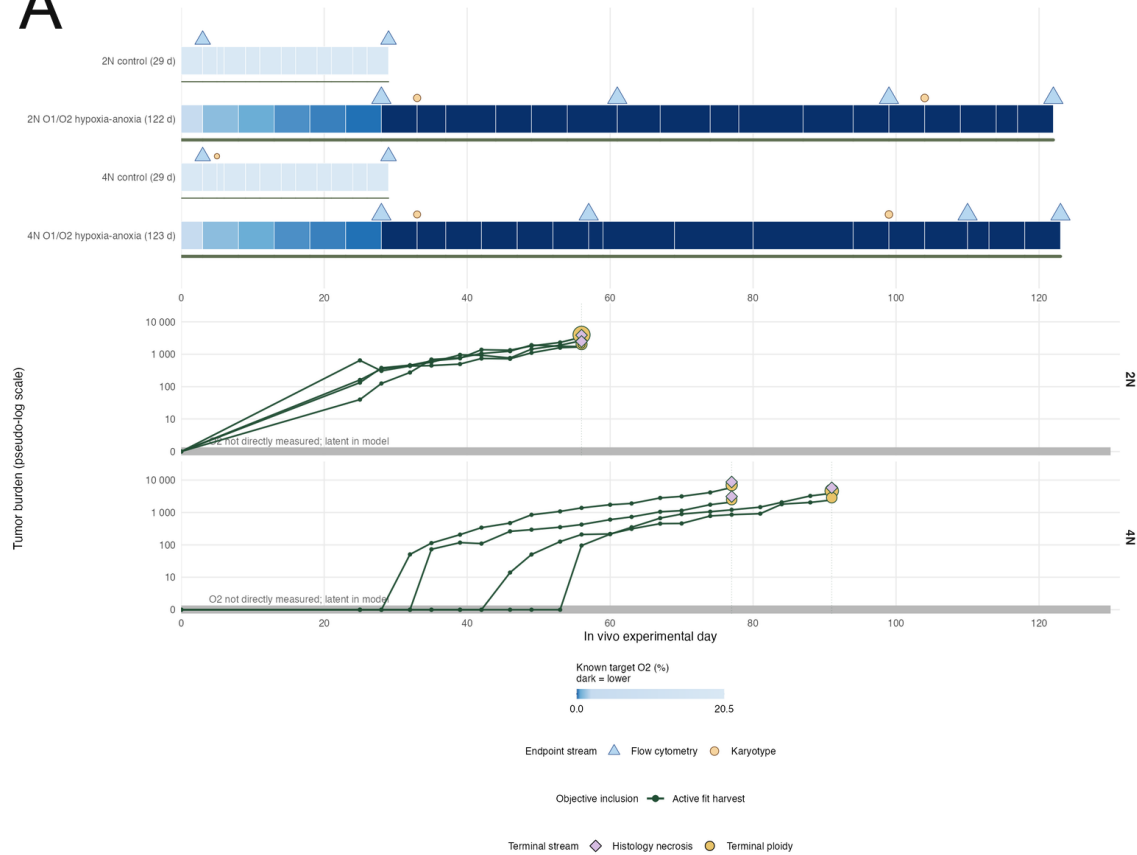


Figure 1: Matched SUM159 lineages reveal environment-dependent ploidy selection.

(A) Overview of the matched SUM159 near-diploid and near-tetraploid lineages analyzed *in vitro* and *in vivo* on comparable experimental time axes. The *in vitro* portion shows control and O1/O2 hypoxia-anoxia passage histories, with target O₂ encoded by blue intensity and measurement events marked for flow cytometry and karyotyping. The *in vivo* portion shows tumor-burden trajectories for the fitted untreated harvest mice, with terminal chromosome-number and histologic necrosis measurements indicated at harvest.

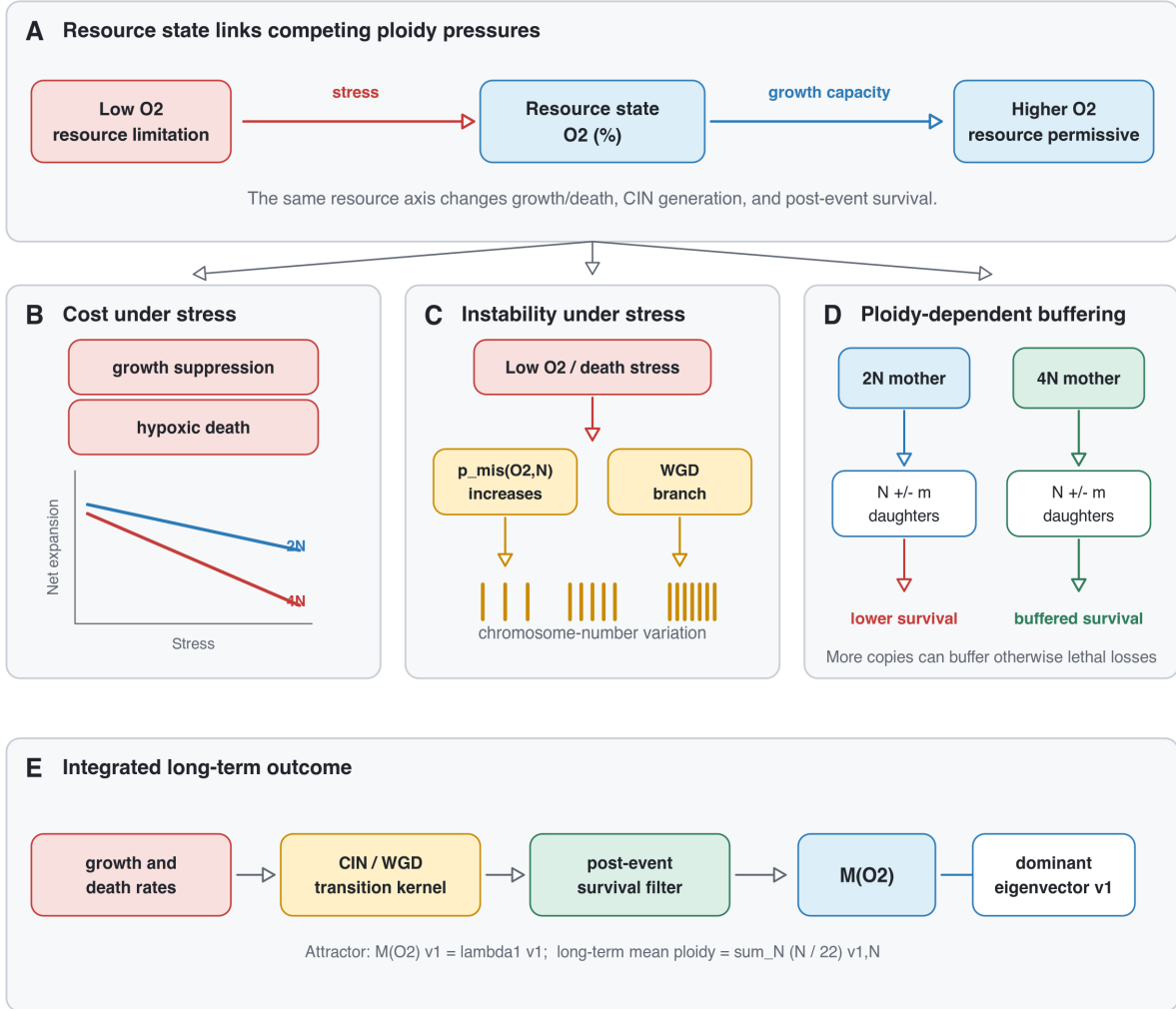


Figure 2: Model overview: resource limitation can impose opposing pressures on ploidy evolution. (A) Resource limitation is represented as a model axis that can reshape ploidy through competing mechanisms. (B) High ploidy is costly under resource stress through growth suppression and death. (C) Stress can increase CIN/WGD pressure, generating chromosome-number variation. (D) High ploidy can buffer otherwise lethal chromosome missegregation. (E) Growth, death, CIN/WGD transitions, and post-missegregation survival define the fixed- O_2 transition matrix; its dominant eigenvector gives the long-term chromosome-state distribution and mean ploidy.

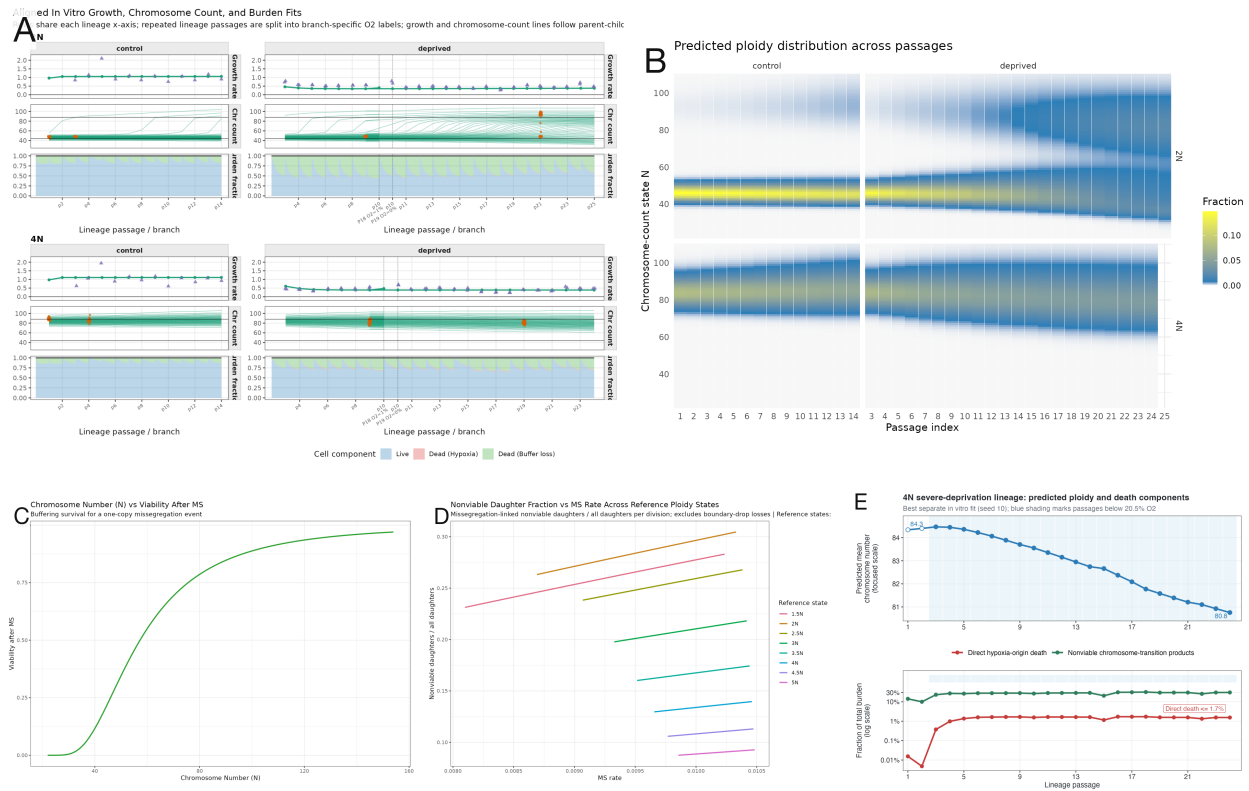


Figure 3: In vitro ploidy reshaping occurs without strong high-ploidy elimination. (A) The separate *in vitro* fit tracks passage growth, chromosome-count observations, and model-predicted live/dead burden components across 2N and 4N control and oxygen-deprived branches. (B) The predicted chromosome-state distribution across passages shows the oxygen-deprived 2N branch moving through a transient high-ploidy/WGD-like expansion, followed by chromosome-loss-derived viable descendants and a shift toward lower-ploidy states. (C) The fitted post-missegregation survival function increases with chromosome number, allowing chromosome-loss daughters from higher-ploidy mothers to remain viable more often than comparable daughters from lower-ploidy mothers. (D) The model-implied nonviable daughter fraction plotted against missegregation rate shows how chromosome-loss opportunity and ploidy-dependent survival jointly determine which chromosome-segregation-error products remain viable across reference ploidy states. (E) Passage-end predictions along the terminal severe-deprivation 4N lineage show a decline in mean chromosome number from 84.3 to 80.8 while model-predicted direct hypoxia-origin dead burden remains at or below 1.72% of total burden. The larger dead-buffer component records nonviable chromosome-transition products, including out-of-grid boundary routing, and is distinct from direct hypoxia-origin killing of the high-ploidy population.

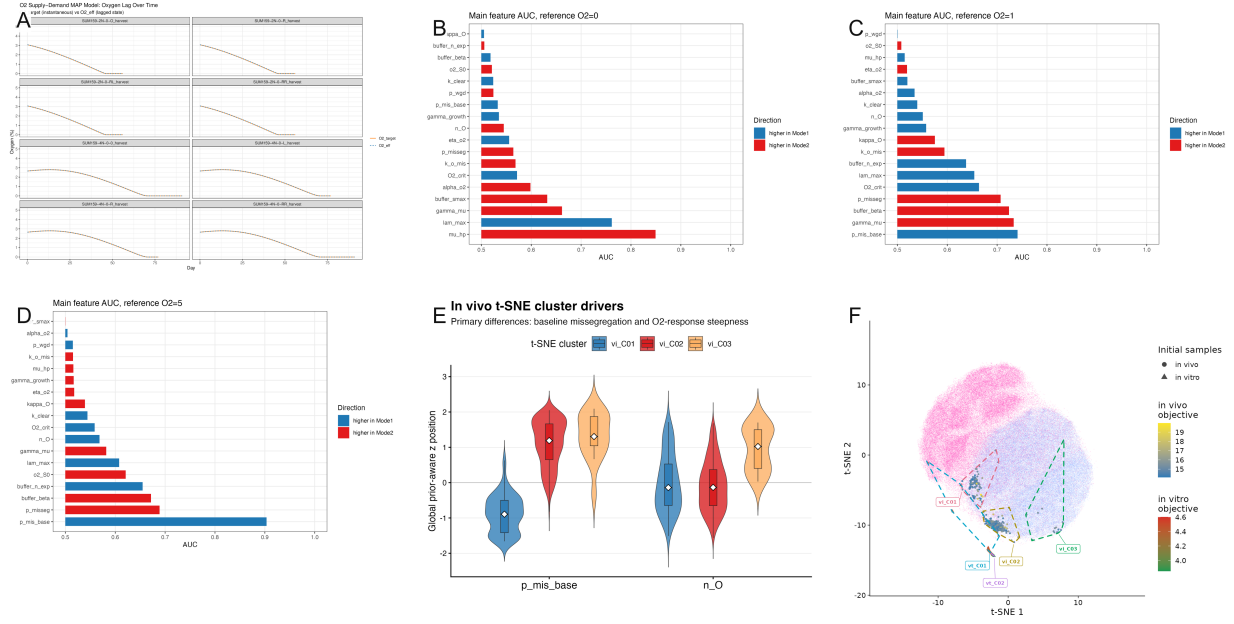


Figure 4: *In vivo* fit-derived resource regimes and parameter landscapes. (A) Fitted effective O₂ trajectories from the separate *in vivo* fit provide a latent model-implied resource axis rather than direct oxygen measurements. (B–D) One-feature AUC screens identify parameters and derived features that best separate lower- from higher-ploidy fixed-O₂ outcomes at low, intermediate, and high reference O₂. (E) Prior-scaled fitted parameter positions for baseline missegregation and oxygen-response shape differ descriptively across three *in vivo* solution clusters. (F) Pooled t-SNE visualization of initial and best-fit *in vivo*/*in vitro* parameter samples, with dashed outlines marking selected solution regions and color scales encoding the corresponding component objective. Panels E–F summarize the fitted parameter landscape and do not establish biological subtypes.

Joint soft-coupled fitting design

Six landscape-informed warm starts; each joint run uses 500 optimizer seeds

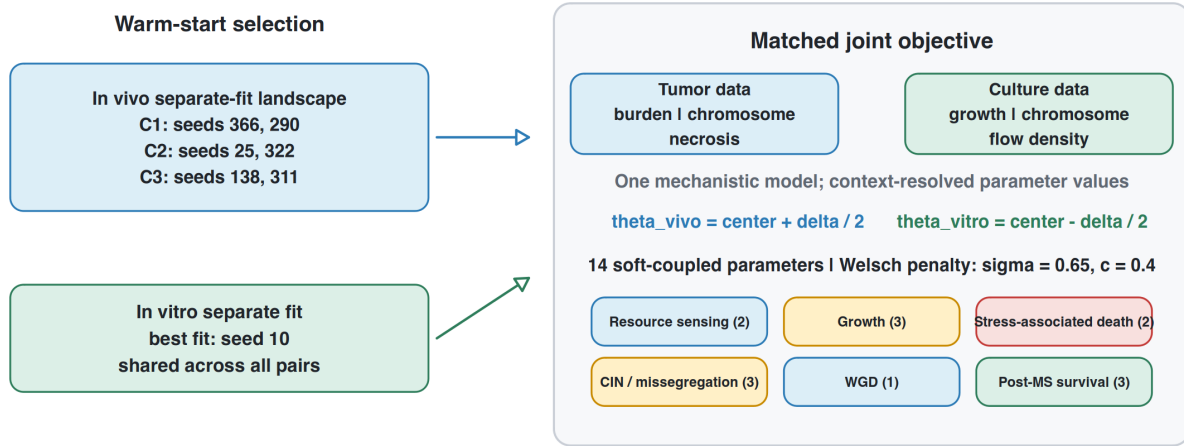


Figure 5: **Joint fits distinguish model-supported context differences.** (A) Joint-fit design. Six *in vivo* landscape-informed warm starts, spanning three clusters and two subclusters per cluster, were paired with the same best separate *in vitro* fit (seed 10); each pair was optimized from 500 numerical seeds. The matched tumor and culture likelihoods used one mechanistic model, while all 14 overlapping active biological parameters received context-specific values reconstructed around a common transformed-scale center and regularized with a Welsch soft-coupling penalty ($\sigma = 0.65$, $c = 0.4$).

Pair-level in-vivo / in-vitro ratio map

Each cell is the median across 500 fitting seeds; pair labels include the pair-level ploidy category

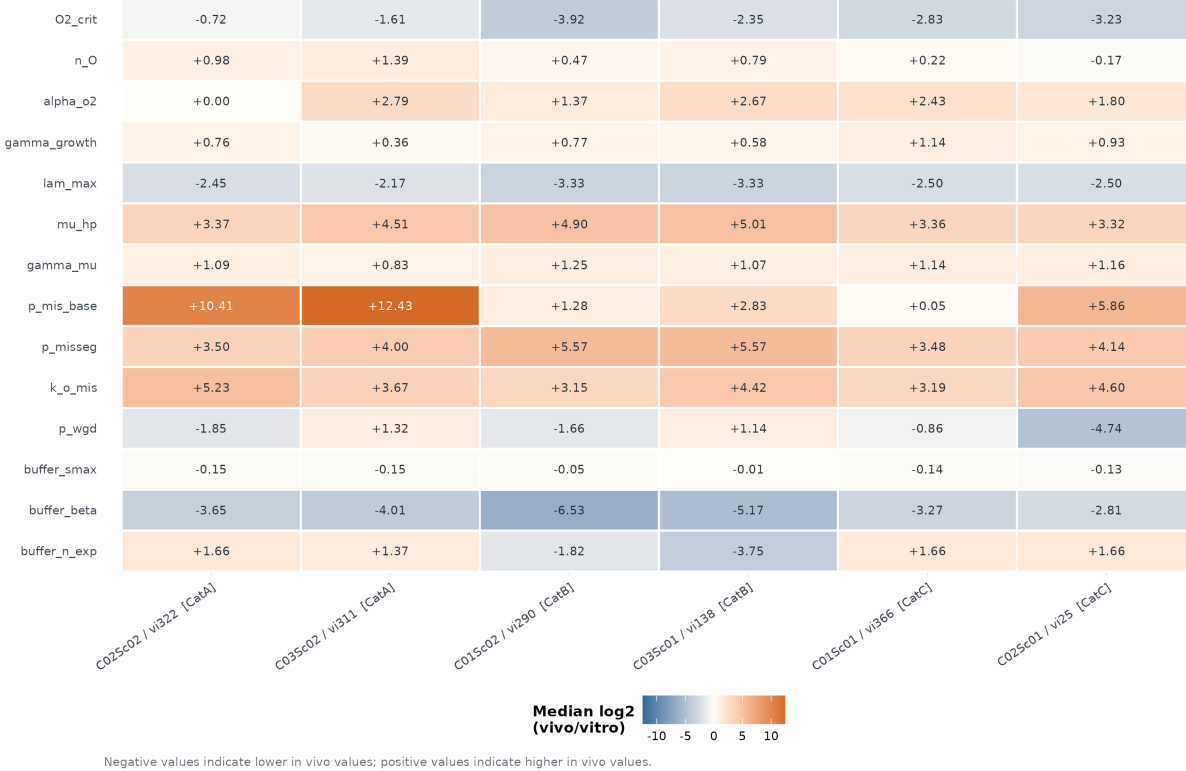


Figure 5: **Joint fits distinguish model-supported context differences (continued).** (B) Pair-level parameter-ratio map. Each cell is the median natural-scale *in vivo*/*in vitro* ratio across the 500 numerical seeds for one warm-start pair. Colors encode $\log_2$ ratios; the response-class annotations identify the separate-fit fixed- O_2 curve category associated with each *in vivo* warm start. Because every pair shares the same *in vitro* seed-10 anchor, these pair labels are descriptive and do not define independent experimental groups.

Between-pair class consensus

Pair is the statistical unit; labels show the exact number out of six pairs

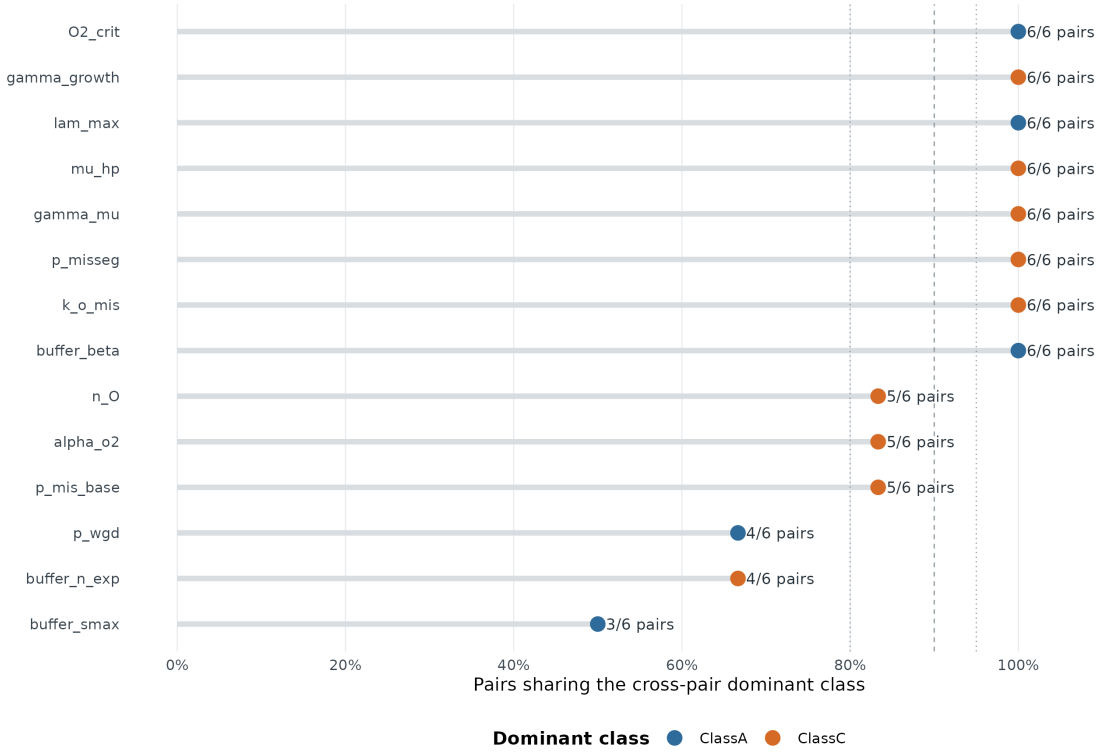


Figure 5: **Joint fits distinguish model-supported context differences (continued).** (C) Directional consensus across the six warm-start pairs. For each parameter, pair-median ratios were classified as lower *in vivo* (ratio $< 1/1.1$), approximately coupled ($1/1.1$ to 1.1), or higher *in vivo* (ratio > 1.1). Fractions therefore use the six warm-start pairs—not the 3,000 optimizer runs or biological specimens—as the summarization unit.

Pair-specific parameter ratio intervals

Gold band marks ClassB; intervals expose both direction and seed-level spread

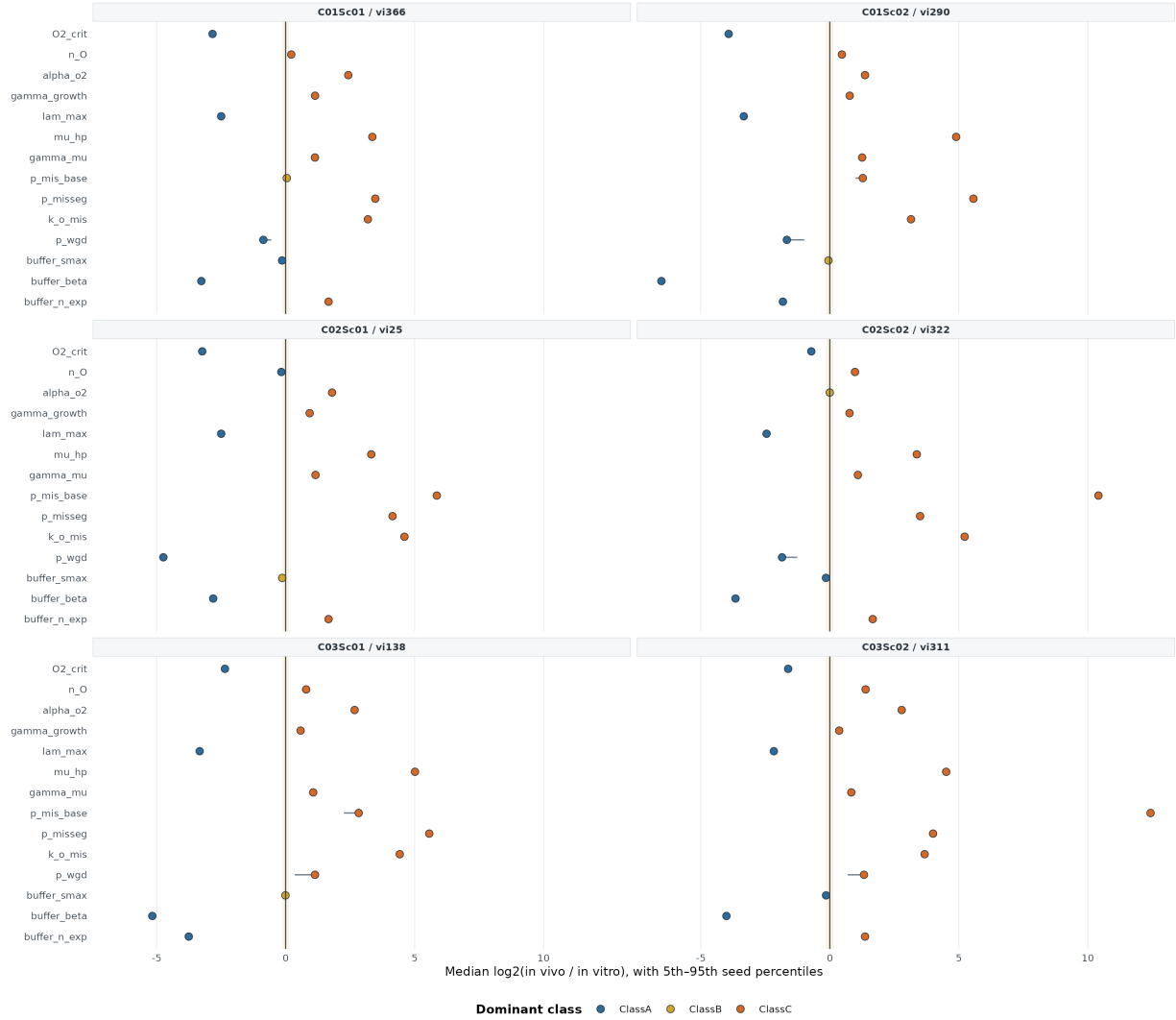


Figure 5: **Joint fits distinguish model-supported context differences (continued).** (D) Pair-specific parameter ratios. Points show medians and horizontal intervals show the 5th–95th percentiles across 500 optimizer seeds within each warm-start pair. These intervals quantify numerical optimizer-start spread; they are not confidence intervals and the seeds are not biological replicates.

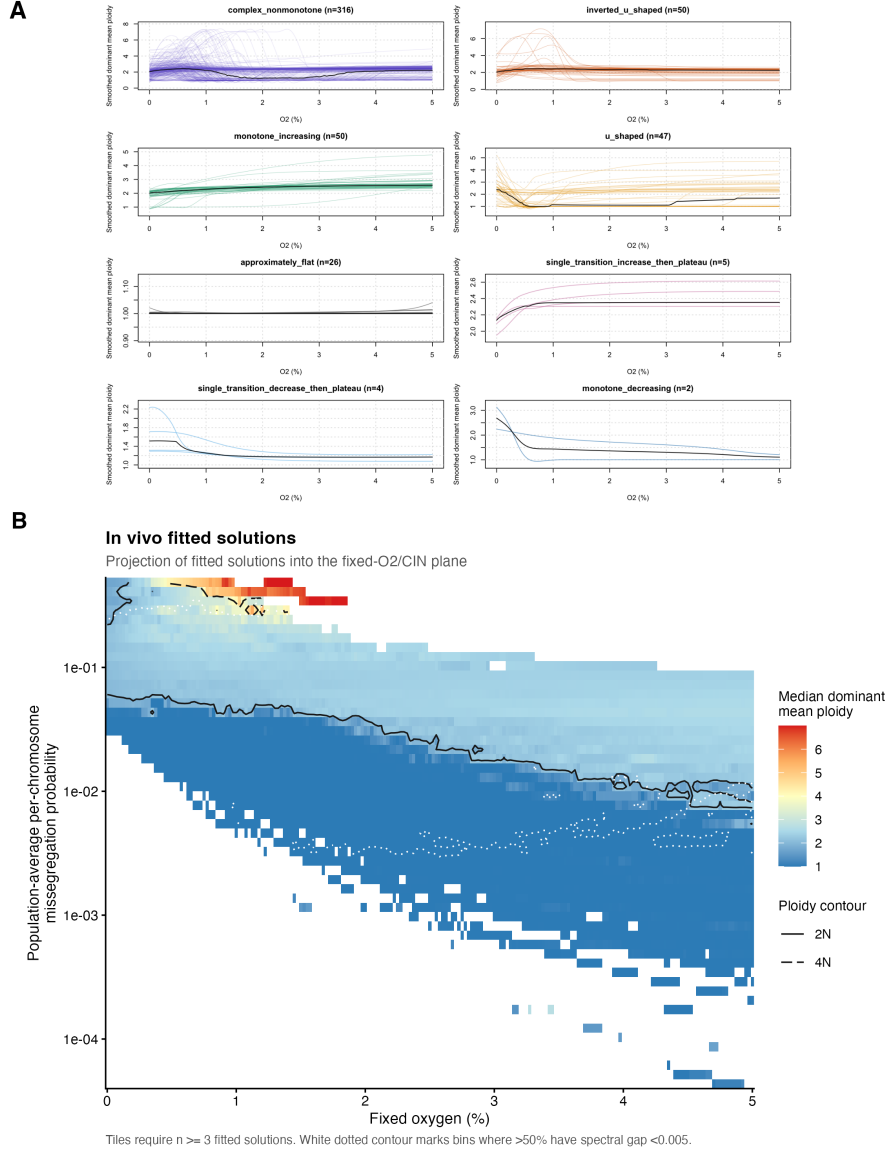


Figure 6: **Fixed-O₂ fits occupy distinct oxygen-CIN-ploidy regimes.** (A) Analytical dominant-eigenvector ploidy curves from 500 separate *in vivo* fits were grouped by regression-smoothed response shape; class sizes are shown above each panel, individual fits are colored, and the black curve summarizes each class. (B) For each seed and fixed-O₂ value, the dominant eigenvector v_N was used to calculate the population-average per-chromosome missegregation probability, $\sum_N v_N p_{\text{mis}}(O_2, N)$. Occupied bins containing at least three fitted solutions are colored by median dominant mean ploidy; solid and dashed black contours mark 2N and 4N, respectively. The white dotted contour marks bins in which more than half of solutions have a spectral gap below 0.005, and unsupported bins are left blank. This panel is a projection of fitted solutions, not an independent oxygen-by-CIN intervention surface.

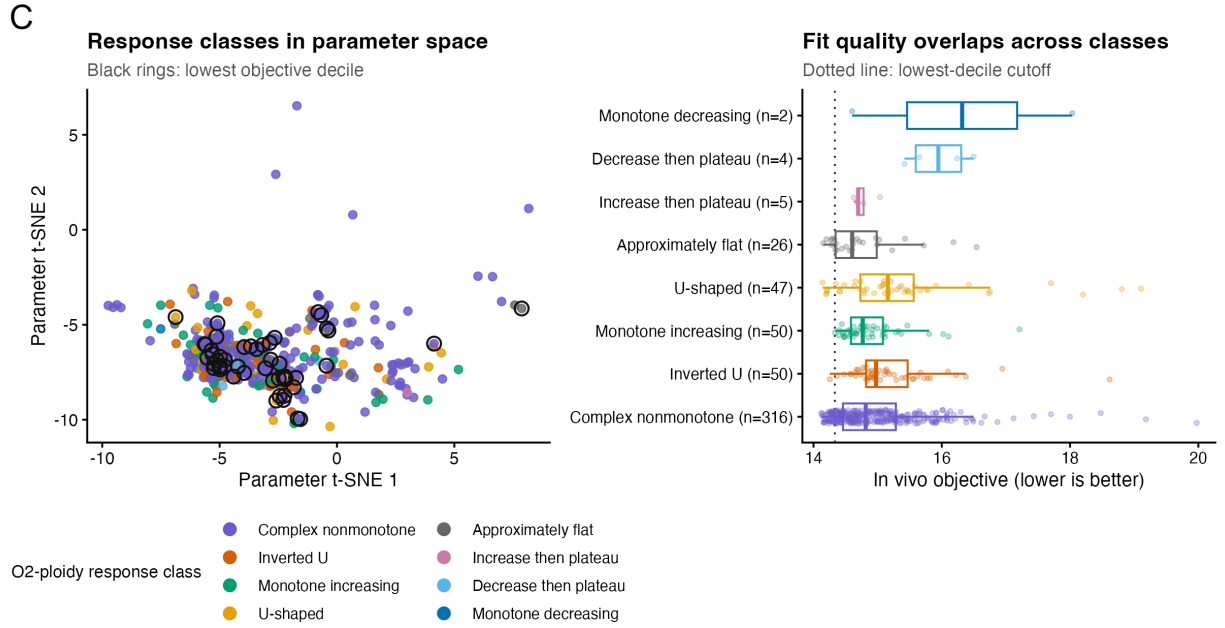


Figure 6: **Fixed- O_2 fits occupy distinct oxygen–CIN–ploidy regimes (continued).** (C) Left, the 500 separate *in vivo* fits are positioned in a t-SNE embedding of prior-scaled fitted parameters and colored by fixed- O_2 response class; black rings mark the lowest 10% of objective values. Right, objective distributions are shown for all response classes, with the dotted line marking the same lowest-decile cutoff. Several common response classes and high-quality fits overlap in parameter space and objective value, and no single response class uniquely contains the best-fitting solutions. The overlay diagnoses remaining model uncertainty rather than selecting one curve class a priori.

Discussion

Here, we integrated matched *in vitro* and *in vivo* ploidy-evolution datasets into a common mechanistic framework. Soft coupling is useful here as a hypothesis-generating regularization strategy: it permits context-specific values while discouraging separation on the transformed parameter scale. The reported Welsch penalty is bounded, however, so sufficiently large differences contribute a saturated penalty rather than an indefinitely increasing cost. Repeated separation across warm-start pairs therefore identifies model-supported context contrasts conditional on the selected likelihoods, bounds, regularization, and common *in vitro* anchor; it is not an equality test and does not by itself establish that the environment caused the difference.

Two contrasts were especially consistent across all six warm-start pairs. The fitted maximal division rate, $\lambda_{\max}$, was lower *in vivo*, which is compatible with a broader tumor-associated growth constraint not represented by oxygen alone. The stress-linked missegregation parameter, p_{misseg} , was higher *in vivo*, although its biological effect is jointly determined with the oxygen response and other missegregation terms, and two pair-specific maxima reached the active upper bound. The fitted oxygen-response threshold was lower and several stress-associated death parameters were higher *in vivo*; these are contrasts in latent model parameters, not direct measurements of tissue oxygen or death mechanisms, and several axes were limited by active parameter bounds. By contrast, the post-missegregation survival function did not show one uniform cross-context shift: β_{buf} was lower in all pairs, $s_{\max}$ alternated between lower and approximately coupled regimes, and n_{exp} was higher in four pairs but lower in two. Because the six pairs share the same *in vitro* seed-10 solution and represent alternative *in vivo* parameter regimes rather than biological replicates, this heterogeneity should be treated as unresolved structural uncertainty. The joint analysis therefore prioritizes proliferative capacity and stress-linked CIN as robust model-generated hypotheses while leaving the composite survival-filter contrast open.

Several observations highlight limitations of a purely resource-dependent interpretation of ploidy evolution. First, aneuploidy has been reported to correlate inversely with growth rate even in energy-rich cell-culture environments. This suggests that abnormal chromosome number can impose a fitness cost even when oxygen and nutrients are not strongly limiting. Such costs may arise from proteotoxic stress, replication stress, mitotic delays, altered gene dosage, or other consequences of karyotypic imbalance that persist under standard culture conditions. Therefore, while resource limitation may amplify selection against high-ploidy states, some component of the high-ploidy or aneuploidy-associated growth penalty may be present even under normoxic, nutrient-replete conditions. This raises the possibility that models in which chromosome-number penalties vanish entirely under low-stress conditions may be too permissive, and that a small baseline ploidy-dependent cost could eventually be needed.

Second, oxygen is only one component of the broader resource environment experienced by tumor cells. Tissue energy availability depends on nutrient concentration, vascular structure, endothelial transport, tight junction permeability, distance from vessels, local consumption by neighboring cells, cell density, and metabolic specialization. Thus, the same nominal oxygen concentration may correspond to very different energetic states across tissue sites or between *in vitro* and *in vivo* settings. In particular, *in vitro* anoxia in glucose-rich medium may not be equivalent to *in vivo* hypoxia, where low oxygen often reflects broader perfusion failure and co-limitation by glucose and other substrates. This motivates treating oxygen as an informative but incomplete proxy

for resource stress, and allowing context-specific differences in how oxygen availability maps onto proliferation, death, and chromosomal instability.

Third, tissue-specific differences in ploidy need not arise solely from resource-mediated selection. Alternative explanations include differences in the rate at which WGD is generated, physiological polyploidization programs in the tissue of origin, immune-mediated selection against high-ploidy or aneuploid cells, and tissue-specific mutagenic pressures. These mechanisms could alter either the production of high-ploidy states or their subsequent survival. For example, tissues with higher baseline polyploidization or mitotic-error rates may generate WGD-positive cells more frequently, whereas immune pressure or mutagenic stress may change the relative advantage of genome doubling. Therefore, failure of a resource-centered model to explain a particular dataset should not necessarily falsify the energetic hypothesis, but may indicate that additional axes of tissue-specific WGD generation, immune selection, or baseline mutagenicity need to be incorporated.

These findings also have implications for therapies that induce chromosome-segregation errors. Many anticancer agents elevate chromosome mis-segregation either directly, by disrupting mitosis, or indirectly, by producing replication and DNA-damage lesions that persist into mitosis.^{17–23} Under the ploidy-dependent survival framework developed here, such agents should not have uniform effects across chromosome-number states: near-diploid cells are predicted to be more vulnerable to missegregation-induced dosage imbalance, whereas WGD or high-ploidy cells may better buffer individual chromosome gains or losses and convert some errors into heritable karyotype variation. This provides a mechanistic interpretation for prior drug-response analyses in which multiple mitotic spindle poisons, including paclitaxel, vinblastine, and vinorelbine, were among the strongest ploidy-associated agents.²⁴ It also suggests why gemcitabine, despite being classically viewed as an S-phase antimetabolite, may show a strong association with ploidy: independent chromosome-loss assays have identified gemcitabine as a potent inducer of chromosome instability.¹⁷ Thus, replication-stress-induced chromosome mis-segregation may create a ploidy-dependent therapeutic vulnerability, linking the model’s survival-filter mechanism to clinically used cytotoxic agents.

1 Supplementary Information

1.1 *In vivo* experiments

The *in vivo* analysis used near-diploid and near-tetraploid variants of the SUM159 triple-negative breast cancer cell line, denoted SUM159_NLS_2N_A6M and SUM159_NLS_4N_A4M, respectively. For each tumor, 1×10^6 cells were injected orthotopically into mice. The subset analyzed here comprised eight tumors: four initiated from the near-diploid 2N line and four initiated from the near-tetraploid 4N line.

- **2N cohort:** four mice injected with SUM159_NLS_2N_A6M.
- **4N cohort:** four mice injected with SUM159_NLS_4N_A4M.

Tumors were harvested at predefined endpoints, and single-cell RNA sequencing was performed on all eight tumors. Chromosome-number distributions used for model calibration were derived from these endpoint single-cell profiles and analyzed together with the corresponding tumor-burden

trajectories. Only this untreated subset was used for the modeling analyses presented here. Processed data and associated experimental metadata were organized and retrieved through CLONEID, a framework for structuring clone-specific genotypic and phenotypic measurements.²⁵

1.2 *In vitro* experiments

The *in vitro* analysis used matched near-diploid and near-tetraploid SUM159 lineages. For each starting ploidy, a control branch was maintained at 20.5% O₂, and two parallel oxygen-deprivation lineages (O1 and O2) were stepped over successive passages through target levels of 2%, 1%, 0.5%, 0.3%, 0.2%, 0.1%, and 0% O₂. The main deprivation branch was then maintained at target 0% O₂; an additional branch after the first 0% passage was evaluated at target 1% O₂. We describe O1 and O2 as parallel lineages because the available records do not establish whether they should be classified as biological replicates.

For calibration, each fit entry retained the assigned oxygen level, passage duration, initial and final corrected cell counts, and a segment-level growth-rate estimate. Chromosome-number measurements and G0/G1 DNA-content density profiles were linked to the corresponding passage when available. The current fit objects contain 131 entries, including 114 with finite growth estimates, 12 with chromosome-number measurements (220 single-cell values in total), and 20 with matched G0/G1 profiles. Each G0/G1 profile was represented on a 200-point ploidy grid and normalized before comparison with the simulated viable chromosome-number distribution.

Within the passage-structured model, oxygen was held fixed at the assigned value for each segment. When a simulated segment was represented by parallel O1 and O2 records, passage duration and initial/final cell counts were summarized by the median, while all available growth, chromosome-number, and flow-density observations contributed to their corresponding likelihood terms.

1.3 A Mechanistic Model of Chromosome-Number Evolution

We model tumor population dynamics as a deterministic state-structured ODE system over chromosome-number classes. The formulation separates viable cells from non-viable biomass, couples local oxygen availability to resource-dependent growth and stress-associated death, and uses chromosome missegregation and whole-genome doubling to move viable mass across chromosome-number states.

State space and compartments. We consider tumor cells stratified by total chromosome count. Let

$$\mathcal{N} = \{N_{\min}, \dots, N_{\max}\}. \quad (1)$$

This set denotes the allowed chromosome-count grid. For the simulations described here, $N_{\min} = 22$ and $N_{\max} = 154$. Let $N_{\text{dip}} = 44$ denote the diploid chromosome-count reference state, and let $N_{\text{unit}} = 22$ denote the one-ploidy chromosome-count unit, so that $2N_{\text{unit}} = N_{\text{dip}}$.

The viable abundance vector is

$$\mathbf{x}^L(t) = \{x_N^L(t)\}_{N \in \mathcal{N}}, \quad (2)$$

where $x_N^L(t)$ is the number of viable cells with chromosome count N at time t . The ODE state is indexed only by chromosome count and by live/dead compartment. Unless explicitly marked as

a dead compartment, chromosome-number distributions in the model are defined over the viable population. The total viable population size is

$$N_{\text{tot}}(t) = \sum_{N \in \mathcal{N}} x_N^L(t). \quad (3)$$

To represent non-viable biomass contributing to macroscopic tumor volume, we track two dead pools:

$$\mathbf{x}^{D,\text{hyp}}(t) = \{x_N^{D,\text{hyp}}(t)\}_{N \in \mathcal{N}}, \quad \mathbf{x}^{D,\text{CIN}}(t) = \{x_N^{D,\text{CIN}}(t)\}_{N \in \mathcal{N}}. \quad (4)$$

Here $\mathbf{x}^{D,\text{hyp}}$ receives cells lost through stress-associated death, and $\mathbf{x}^{D,\text{CIN}}$ receives non-viable daughter mass generated by chromosome missegregation or WGD; the superscript CIN denotes missegregation-origin loss or out of grid WGD. Their sum is

$$\mathbf{x}^D(t) = \mathbf{x}^{D,\text{hyp}}(t) + \mathbf{x}^{D,\text{CIN}}(t). \quad (5)$$

The model's bulk tumor burden is therefore

$$N_{\text{burden}}(t) = \sum_{N \in \mathcal{N}} \left(x_N^L(t) + x_N^{D,\text{hyp}}(t) + x_N^{D,\text{CIN}}(t) \right), \quad (6)$$

which includes viable cells and dead biomass. This distinction is used throughout: tumor-volume observations are compared with total burden, whereas single-cell chromosome-number observations sample the viable distribution.

Environmental stress functions. Let $O_2(t)$ denote the mean-field approximation of the effective oxygen level experienced by the tumor population. Oxygen/resource stress is represented by the bounded Hill function

$$h(O_2) = \frac{O_{2,c}^{n_O}}{O_{2,c}^{n_O} + O_2^{n_O}}, \quad (7)$$

where $O_{2,c}$ is the critical oxygen level at which the stress term is half-maximal, and n_O controls the steepness of the transition from low stress at high oxygen to strong stress under oxygen limitation. A Hill-type stress function is biologically justified because the literature indicates that oxygen-dependent cell fate is graded but threshold-like rather than linear: “The severity of hypoxia determines whether cells become apoptotic or adapt to hypoxia and survive”,²⁶ and recent spheroid work identifies a “critical survival pO_2 ”;²⁷ accordingly, Eq. (7) provides a smooth, bounded approximation to an oxygen-linked resource-stress threshold.

Proliferation and stress-associated death rates. Given chromosome count N and local oxygen level O_2 , the effective division rate is

$$\lambda_{\text{eff}}(N, O_2) = \lambda_{\text{max}} \frac{1}{1 + \alpha_{O_2} h(O_2) \left(\frac{N}{N_{\text{dip}}} \right)^{\gamma_{\text{growth}}}}. \quad (8)$$

Here α_{O_2} controls the strength of resource-stress-dependent growth damping, and γ_{growth} controls how strongly this damping increases with chromosome content. When oxygen-linked stress is small, $h(O_2) \approx 0$ and the reciprocal factor approaches one; as oxygen decreases, the reciprocal factor

suppresses proliferation more strongly, especially for high-chromosome-count states.

Stress-associated death is applied as an explicit live-to-dead flow. The per-capita death hazard is

$$\mu_{\text{eff}}(N, O_2) = \mu_{\text{hp}} h(O_2) \left(\frac{N}{N_{\text{dip}}} \right)^{\gamma_\mu}. \quad (9)$$

Thus μ_{hp} sets the stress-associated death scale, $h(O_2)$ makes death depend on the same oxygen-linked stress function (Eq. (7)), and γ_μ controls the chromosome-number penalty on stress-associated death.

Cell division, WGD, and chromosomal transitions. With the stress-modulated rates defined, divisions are represented by a live-state generator matrix

$$G_{\text{div}}(O_2) \in \mathbb{R}^{R \times R}, \quad R = N_{\text{max}} - N_{\text{min}} + 1. \quad (10)$$

Let

$$L(O_2) = \text{diag}(\lambda_{\text{eff}}(N, O_2)) \quad (11)$$

be the diagonal matrix of state-specific division rates. A division follows the ordinary chromosome-missegregation branch with probability $1 - p_{\text{wgd}}$ and follows a whole-genome-doubling branch with probability p_{wgd} . The corresponding division generator is

$$G_{\text{div}}(O_2) = (1 - p_{\text{wgd}}) B(O_2) L(O_2) + p_{\text{wgd}} W L(O_2) - L(O_2), \quad (12)$$

where $B(O_2)$ is the viable offspring-allocation matrix generated by chromosome missegregation and post-missegregation viability filtering, W maps a mother state N to the within-grid doubled state $2N$, and the final $-L(O_2)$ term removes the dividing mother cell. The ordinary branch B accounts for the two daughter branches of a division and therefore has total mass up to two before viability losses. By contrast, the WGD branch is implemented as an effective tracked-lineage conversion: W has unit mass in the doubled state, so a WGD event replaces the mother lineage by one viable doubled lineage when $2N$ lies in the modeled grid. Out-of-grid WGD mass is not retained in the viable state space and is routed to the CIN-associated dead compartment.

Chromosome missegregation and daughter viability. For a mother cell with chromosome count N , the per-chromosome missegregation probability increases with the stress-associated death hazard defined in Eq. (9):

$$p_{\text{mis}}(N, O_2) = p_{\text{mis}, \text{base}} + p_{\text{misseg}} \frac{\mu_{\text{eff}}(N, O_2)}{\mu_{\text{eff}}(N, O_2) + k_{o, \text{mis}}}, \quad (13)$$

with values clipped to $[0, 1]$. Here $p_{\text{mis}, \text{base}}$ is the baseline per-chromosome missegregation probability under low stress, p_{misseg} is the maximal stress-induced increment above baseline, and $k_{o, \text{mis}}$ is the death-hazard scale at which the stress-induced increase in missegregation is half-maximal.

Let m denote the number of missegregated chromosome copies in a division of a mother cell with chromosome count N . We model

$$m \sim \text{Binomial}\{N, p_{\text{mis}}(N, O_2)\}. \quad (14)$$

For each value of m , the two daughter branches are assigned coarse chromosome-count shifts $+m$ and $-m$. Before applying the daughter-viability modifier, the shift kernel is

$$d_N(\Delta; O_2) = \sum_{m=0}^N \binom{N}{m} \{p_{\text{mis}}(N, O_2)\}^m \{1 - p_{\text{mis}}(N, O_2)\}^{N-m} [\mathbf{1}\{\Delta = m\} + \mathbf{1}\{\Delta = -m\}]. \quad (15)$$

This kernel sums over two daughter branches, so its total mass is two before viability filtering.

For a daughter with chromosome-count shift Δ , post-missegregation survival is approximated by the symmetric ploidy-dependent modifier

$$S_N(\Delta) = s_N^{|\Delta|}, \quad s_N = s_{\text{max}} \exp \left[-\beta_{\text{buf}} \left(\frac{N_{\text{dip}}}{N} \right)^{n_{\text{exp}}} \right]. \quad (16)$$

The parameter s_{max} is the maximum per-copy survival factor, β_{buf} controls the severity of post-missegregation viability loss at lower ploidy, and n_{exp} controls the steepness of the ploidy dependence. This form lets higher-chromosome-count states tolerate a given chromosome-count shift more readily than lower-chromosome-count states. This survival modifier enters the retained-grid offspring allocation and dead-pool accounting described in the boundary-handling subsection below.

Boundary handling on the chromosome-count domain. The chromosome-count state space is finite. We define the viable chromosome-count domain as

$$\mathcal{N} = \{N_{\text{min}}, N_{\text{min}} + 1, \dots, N_{\text{max}}\},$$

where $N_{\text{min}} = 22$ and $N_{\text{max}} = 154$ in the simulations described here. The viable state vector

$$\mathbf{x}^L(t) = \{x_N^L(t)\}_{N \in \mathcal{N}}$$

therefore tracks only cells whose total chromosome count lies within this retained grid.

This finite state space requires an explicit rule for division events that would produce daughter or doubled-lineage states outside $\mathcal{N}$. We use absorbing boundary handling on the viable chromosome-count grid. That is, daughter cells or WGD-derived lineages whose chromosome count would fall below N_{min} or exceed N_{max} are removed from the viable offspring allocation and routed to the CIN-associated dead compartment.

For ordinary missegregating divisions, the viable offspring-allocation matrix is defined only over retained daughter states $N' \in \mathcal{N}$:

$$B_{N', N}(O_2) = d_N(N' - N; O_2) S_N(N' - N), \quad N, N' \in \mathcal{N}.$$

Here $d_N(\Delta; O_2)$ specifies the probability of generating a daughter chromosome-count shift Δ , and $S_N(\Delta)$ specifies the probability that a daughter with that shift remains viable. Any probability mass corresponding to $N + \Delta \notin \mathcal{N}$ is excluded from $B_{\cdot, N}$. Similarly, for whole-genome-doubling events, the doubling operator W retains only within-grid transitions $N \mapsto 2N \in \mathcal{N}$; out-of-grid WGD offspring mass is excluded from $W_{\cdot, N}$ and routed to the CIN-associated dead compartment.

Because the viable offspring allocation can be subunitary, either because buffering reduces daughter survival or because daughter or doubled-lineage states fall outside the retained chromosome-

count grid, we define the CIN-associated event-level nonviable inflow as

$$\phi_{\text{CIN},N}(\mathbf{x}^L, O_2) = \lambda_{\text{eff}}(N, O_2) x_N^L \left\{ (1 - p_{\text{wgd}}) \left[2 - \sum_{N' \in \mathcal{N}} B_{N',N}(O_2) \right] + p_{\text{wgd}} \left[1 - \sum_{N' \in \mathcal{N}} W_{N',N} \right] \right\}. \quad (17)$$

The first bracket is the ordinary-division deficit between two potential daughters and the viable daughter mass retained within $\mathcal{N}$. The second bracket accounts for WGD events for which the doubled state $2N$ lies outside the modeled grid. This term is distinct from the continuous hypoxia-associated death hazard $\mu_{\text{eff}}(N, O_2)$; it is added to the CIN-associated dead compartment $x^{D,\text{CIN}}$.

Thus, the chromosome-count boundaries are absorbing for viable offspring mass, with out-of-domain and non-buffered daughter mass transferred to the dead compartment.

Oxygen demand and supply dynamics. To couple tumor size to oxygen demand, we define the chromosome-number-weighted oxygen-demand proxy

$$N_{\text{eff}}(t) = \sum_N x_N^L(t) \left(\frac{N}{N_{\text{dip}}} \right)^\eta, \quad (18)$$

where η controls how oxygen demand scales with chromosome content. The reference scale N_{ref} is a fixed cell-count scale used to nondimensionalize demand, so that $N_{\text{eff}}/N_{\text{ref}}$ is a relative effective tumor size.

The oxygen supply target is the phenomenological supply–demand relation

$$O_2^*(N_{\text{eff}}) = \max \left(O_{2,\text{min}}, S_0 - \kappa_O \log \left(1 + \frac{N_{\text{eff}}}{N_{\text{ref}}} \right) \right), \quad (19)$$

where S_0 is the low-burden oxygen supply level, $O_{2,\text{min}}$ is the oxygen floor, and κ_O controls the oxygen-drop amplitude per unit log-burden increase. The logarithmic term is dimensionless because of the normalization by N_{ref} . Oxygen values are interpreted as percentages and are restricted to the physical interval $[0, 100]$.

Effective oxygen is updated by first-order relaxation toward this target:

$$\alpha_\tau = 1 - \exp \left(-\frac{\Delta t}{\tau_{O_2}} \right), \quad O_{2,k+1} = O_{2,k} + \alpha_\tau (O_2^*(N_{\text{eff},k}) - O_{2,k}). \quad (20)$$

Thus oxygen changes gradually toward the demand-adjusted target rather than instantaneously matching it at each simulation step.

Discrete-time simulation scheme. The simulations use a time step of $\Delta t = 0.05$ days. At each step $k \rightarrow k+1$, the live compartment is updated by division-linked chromosomal transitions and explicit stress-associated death:

$$\mathbf{x}_{k+1}^L = \mathbf{x}_k^L + \Delta t G_{\text{div}}(O_{2,k}) \mathbf{x}_k^L - \Delta t \mu_{\text{eff}}(N, O_{2,k}) \odot \mathbf{x}_k^L, \quad (21)$$

where $\odot$ denotes element-wise multiplication over chromosome-count states.

Mass that becomes nonviable due to hypoxia or failed divisions is transferred into the dead compartments, which contribute to tumor burden. The resource-stress-origin dead pool is updated

as

$$\mathbf{x}_{k+1}^{D,\text{hyp}} = \mathbf{x}_k^{D,\text{hyp}} + \Delta t \mu_{\text{eff}}(N, O_{2,k}) \odot \mathbf{x}_k^L - \Delta t k_{\text{clear}} \mathbf{x}_k^{D,\text{hyp}}, \quad (22)$$

and the CIN-associated dead pool is updated as

$$\mathbf{x}_{k+1}^{D,\text{CIN}} = \mathbf{x}_k^{D,\text{CIN}} + \Delta t \phi_{\text{CIN}}(\mathbf{x}_k^L, O_{2,k}) - \Delta t k_{\text{clear}} \mathbf{x}_k^{D,\text{CIN}}. \quad (23)$$

Here k_{clear} is the first-order clearance rate for dead biomass, and ϕ_{CIN} is defined component-wise in Eq. (17). Oxygen is then updated by Eq. (20). Together, Eqs. (21)–(23) define the coupled viable/dead population dynamics used in the simulations.

1.3.1 Model calibration and observation models

Model calibration uses a common transformed-parameter optimizer for *in vivo*, *in vitro*, and joint fits. Natural-scale parameter specifications provide initialization values and admissible bounds; each component maps its fitted parameters to the optimizer scale before evaluating its objective. The mechanistic dynamics described above are shared, while the observation models differ between the *in vivo* and *in vitro* data.

***In vivo* observation model and likelihood.** The *in vivo* component uses mouse tumor trajectories with two data modalities: tumor burden and ploidy.

Tumor burden. For each mouse $i \in \mathcal{I}$, where $\mathcal{I}$ denotes the eight retained tumors used for calibration (section 1.1), tumor burden is measured at observation times $\{t_{ij}^{(V)}\}_{j=1}^{m_i}$, yielding observed tumor volumes

$$V_{ij}^{\text{obs}} \equiv V_i^{\text{obs}}(t_{ij}^{(V)}). \quad (24)$$

Let $x_{i,N}^L(t)$, $x_{i,N}^{D,\text{hyp}}(t)$, and $x_{i,N}^{D,\text{CIN}}(t)$ denote the viable, hypoxia-origin dead, and CIN-associated dead abundances for tumor i in chromosome-count state N at time t . The model predicts tumor volume by scaling total burden:

$$\hat{V}_{ij} = \sum_N \left(x_{i,N}^L(t_{ij}^{(V)}) + x_{i,N}^{D,\text{hyp}}(t_{ij}^{(V)}) + x_{i,N}^{D,\text{CIN}}(t_{ij}^{(V)}) \right) v_{\text{cell}}, \quad (25)$$

where each cell is assigned the effective volume

$$v_{\text{cell}} = \frac{1}{\rho_{2N}}. \quad (26)$$

This observation model does not impose a ploidy-dependent cell-volume correction: low- and high-ploidy cells contribute the same effective per-cell volume to the bulk burden. Tumor-burden observations are modeled on the logarithmic scale,

$$\log V_{ij}^{\text{obs}} \sim \mathcal{N}\left(\log \hat{V}_{ij}, \sigma_{\text{burden}}^2\right), \quad (27)$$

where σ_{burden} is the burden measurement-noise scale. For longitudinal tumor burden, the fitted loss is a Gaussian negative log-likelihood on the logarithmic scale, averaged first within tumor and then across tumors. Let $\mathcal{J}_i$ denote the retained observation times for tumor i after excluding day

0. Then

$$\mathcal{L}_B(\theta) = \frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} \frac{1}{|\mathcal{J}_i|} \sum_{j \in \mathcal{J}_i} \left[\log \sigma_{\text{burden}} + \frac{1}{2} \log(2\pi) + \frac{(\log V_{ij}^{\text{obs}} - \log \hat{V}_{ij})^2}{2\sigma_{\text{burden}}^2} \right]. \quad (28)$$

Because σ_{burden} is estimated jointly, the $\log \sigma_{\text{burden}}$ term is retained in the likelihood contribution. Omitting it would allow the optimizer to reduce the burden loss by inflating the assumed measurement noise.

Terminal necrosis. For tumors with matched terminal histology, necrosis is compared with the simulated terminal fraction of total dead biomass. This is an observation-model term, not an additional live-to-dead flux. The stress-associated death hazard $\mu_{\text{eff}}(N, O_2)$ contributes to $\mathbf{x}^{D, \text{hyp}}$, while mitotic nonviability and boundary-dropped offspring contribute to $\mathbf{x}^{D, \text{CIN}}$. The terminal necrosis comparison uses the sum of these two dead pools.

For tumor i , define the terminal total-dead volume and terminal total tumor volume as

$$\hat{V}_i^{D, \text{tot}}(\theta) = v_{\text{cell}} \sum_N \left[x_{i,N}^{D, \text{hyp}}(T_i^{(H)}; \theta) + x_{i,N}^{D, \text{CIN}}(T_i^{(H)}; \theta) \right], \quad \hat{V}_i^{\text{tot}}(\theta) = v_{\text{cell}} \sum_N \left[x_{i,N}^L(T_i^{(H)}; \theta) + x_{i,N}^{D, \text{hyp}}(T_i^{(H)}; \theta) + \right. \quad (29)$$

The predicted terminal necrotic fraction is

$$\hat{f}_i^{\text{nec}}(\theta) = \frac{\hat{V}_i^{D, \text{tot}}(\theta)}{\hat{V}_i^{\text{tot}}(\theta)}. \quad (30)$$

Because the same effective cell-volume factor appears in both numerator and denominator, it cancels in this ratio. Thus the fitted necrosis term constrains the terminal total-dead fraction of the simulated tumor burden, after both dead pools have accumulated their respective inflows and have been cleared by the shared first-order clearance process.

Let $f_i^{\text{nec, obs}}$ be the observed terminal necrotic fraction for tumors in the matched necrosis set $\mathcal{I}_{\text{nec}}$. Fractions are compared on a clipped logit scale. Define

$$\text{logit}_\epsilon(f) = \log \left(\frac{\tilde{f}}{1 - \tilde{f}} \right), \quad \tilde{f} = \min\{1 - \epsilon_{\text{nec}}, \max(\epsilon_{\text{nec}}, f)\}. \quad (31)$$

The terminal necrosis loss is the mean standardized squared discrepancy on this scale,

$$\mathcal{L}_{\text{nec}}(\theta) = \frac{1}{|\mathcal{I}_{\text{nec}}|} \sum_{i \in \mathcal{I}_{\text{nec}}} \frac{1}{2} \left[\frac{\text{logit}_\epsilon(\hat{f}_i^{\text{nec}}(\theta)) - \text{logit}_\epsilon(f_i^{\text{nec, obs}})}{\sigma_{\text{nec}}} \right]^2. \quad (32)$$

Here σ_{nec} sets the tolerated discrepancy in terminal necrosis on the logit-fraction scale, and ϵ_{nec} prevents fractions at 0 or 1 from producing infinite logits.

Ploidy. Each mouse has an initial ploidy group $g_i \in \{2N, 4N\}$ and harvest time $T_i^{(H)}$. Retained burden observations are nonzero-time measurements up to harvest, so the number and timing of observations may differ across mice.

For each tumor $i \in \mathcal{I}$, endpoint single-cell copy-number profiling provides total chromosome-number estimates. If n_i cells are profiled from tumor i , the observed chromosome-number values

are denoted

$$z_{i\ell}, \quad \ell = 1, \dots, n_i. \quad (33)$$

These single-cell observations sample the viable population at harvest. The model-predicted viable chromosome-number distribution at harvest is

$$\hat{\pi}_{i,N}(\theta) = \frac{x_{i,N}^L(T_i^{(H)}; \theta)}{\sum_{N'} x_{i,N'}^L(T_i^{(H)}; \theta)}. \quad (34)$$

This distribution represents the predicted probability that a viable cell sampled from tumor i at harvest has true chromosome count N .

Observed single-cell chromosome-number estimates are then modeled conditional on this predicted viable distribution. Specifically, each observed value $z_{i\ell}$ is treated as a noisy measurement of an unobserved true chromosome-count state N , with mixture weights given by $\hat{\pi}_{i,N}(\theta)$:

$$p(z_{i\ell} \mid \theta) = \sum_N \hat{\pi}_{i,N}(\theta) \mathcal{N}(z_{i\ell}; N, \sigma_{\text{ploidy}}^2). \quad (35)$$

where σ_{ploidy} represents measurement variability in single-cell chromosome-number estimates. For terminal single-cell chromosome-number data, the fitted loss first computes a tumor-level average negative log-likelihood and then aggregates these losses with equal weight across tumors. Define

$$\ell_i(\theta) = -\frac{1}{n_i} \sum_{\ell=1}^{n_i} \log p(z_{i\ell} \mid \theta). \quad (36)$$

The endpoint contribution is

$$\mathcal{L}_C(\theta) = \frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} \ell_i(\theta). \quad (37)$$

Maximum-a-posteriori objective. The *in vivo* component is formulated as a maximum-a-posteriori objective combining burden likelihood, chromosome-number likelihood, terminal necrosis loss, and soft-prior penalties:

$$\mathcal{L}_{\text{vivo}}(\theta) = \mathcal{L}_B(\theta) + \mathcal{L}_C(\theta) + \lambda_{\text{nec}} \mathcal{L}_{\text{nec}}(\theta) + \lambda_{\text{prior}} \mathcal{L}_{\text{prior}}(\theta), \quad (38)$$

where θ denotes the fitted parameter vector for this component, $\mathcal{L}_B$ is the burden contribution, $\mathcal{L}_C$ is the terminal chromosome-number contribution, $\mathcal{L}_{\text{nec}}$ is the terminal total-dead-fraction loss, λ_{nec} is its relative objective weight, and $\mathcal{L}_{\text{prior}}$ is the soft-prior penalty.

The soft-prior penalty is

$$\mathcal{L}_{\text{prior}}(\theta) = \frac{1}{2} \sum_{i \in \mathcal{P}} \left(\frac{\theta_i - \mu_i}{\sigma_i} \right)^2, \quad (39)$$

where $\mathcal{P}$ is the subset of parameters with literature-informed priors and each θ_i is assigned an independent Gaussian prior $\theta_i \sim \mathcal{N}(\mu_i, \sigma_i^2)$.

***In vitro* observation model and likelihood.** The *in vitro* component uses the same chromosome-number transition model but applies it to passage-structured cell-line data. Each passage segment

s has an initial cohort label, a fixed oxygen level during the segment, and a duration Δt_s ; available growth observations are represented as segment-level growth-rate data rather than being restricted to a single initial–final cell-count pair. The model propagates the corresponding viable chromosome-number distribution through the segment and compares the simulated passage outcome with three measured data modalities: passage growth-rate observations, passage-level chromosome-number observations, and flow-density measurements. For each passage, the propagated chromosome-number distribution used for reseeding and endpoint comparison was taken from the simulated time point whose live-cell count was closest to the observed final or next-passage initial cell count.

For growth-rate data, let g_s^{obs} be the observed passage growth rate and $\hat{g}_s(\theta)$ the model-predicted growth rate for segment s . The averaged growth log-likelihood is

$$\bar{\ell}_{\text{growth}}(\theta) = \frac{1}{|\mathcal{S}_g|} \sum_{s \in \mathcal{S}_g} \log \mathcal{N}(g_s^{\text{obs}}; \hat{g}_s(\theta), \sigma_{\text{growth}}^2), \quad (40)$$

where $\mathcal{S}_g$ is the set of segments with observed growth-rate data and σ_{growth} is the growth measurement-noise scale.

For chromosome-number observations in segment s , let $\pi_{s,N}$ denote the predicted viable chromosome-number distribution at the comparison time and let $z_{s\ell}$ be the observed chromosome-number value for cell ℓ . The averaged chromosome-number log-likelihood is

$$\bar{\ell}_{\text{ploidy}}(\theta) = \frac{1}{|\mathcal{S}_z|} \sum_{s \in \mathcal{S}_z} \frac{1}{n_s} \sum_{\ell=1}^{n_s} \log \left[\sum_N \pi_{s,N} \mathcal{N}(z_{s\ell}; N, \sigma_{\text{kary}}^2) \right], \quad (41)$$

where $\mathcal{S}_z$ is the set of passage comparisons with chromosome-number observations, n_s is the number of observations in segment s , and σ_{kary} is the chromosome-number measurement-noise scale for the *in vitro* karyotype term.

For flow-density data, let ω_{sp}^{obs} be the normalized observed density mass at flow grid point p for segment s , and let $\hat{f}_s(p; \theta)$ be the model-predicted density after smoothing the simulated viable distribution onto the observed flow grid. The averaged flow log-likelihood is

$$\bar{\ell}_{\text{flow}}(\theta) = \frac{1}{|\mathcal{S}_f|} \sum_{s \in \mathcal{S}_f} \sum_{p \in \mathcal{G}_s} \omega_{sp}^{\text{obs}} \log \hat{f}_s(p; \theta), \quad (42)$$

where $\mathcal{S}_f$ is the set of flow-profile comparisons and $\mathcal{G}_s$ is the observed flow grid for segment s .

The fitted *in vitro* objective is the negative weighted mean log-likelihood

$$\mathcal{L}_{\text{vitro}}(\theta) = - [w_{\text{growth}} \bar{\ell}_{\text{growth}}(\theta) + w_{\text{ploidy}} \bar{\ell}_{\text{ploidy}}(\theta) + w_{\text{flow}} \bar{\ell}_{\text{flow}}(\theta)]. \quad (43)$$

The three terms compare the simulated passage lineages with growth-rate, chromosome-number, and flow-density data, respectively.

Joint fitting of *in vivo* and *in vitro* data. The joint fit uses a single optimizer vector while evaluating separate transformed parameter vectors for the *in vivo* and *in vitro* likelihood components. Soft-coupled parameters are represented by a center–delta parameterization: for each soft-coupled parameter, the optimizer carries one transformed center variable, c_Q , and one addi-

tional offset variable, δ_Q . The context-specific transformed values are reconstructed from these two optimizer variables during objective evaluation. Parameters that appear only in one observation model are retained as context-specific observation-model parameters. In the analysis reported here, all 14 overlapping active biological parameters— $O_{2,c}$, μ_{hp} , p_{misseg} , $k_{o,mis}$, s_{max} , β_{buf} , n_{exp} , n_O , α_{O_2} , γ_{growth} , λ_{max} , $p_{mis,base}$, p_{wgd} , and γ_μ —were soft-coupled across contexts; no overlapping active biological parameter was hard-shared. Thus, the strength and chromosome-number dependence of resource-stress growth suppression were allowed to differ between the *in vivo* and *in vitro* components, while regularization discouraged unsupported divergence.

The minimized joint objective is

$$\mathcal{L}_{\text{joint}}(\theta) = w_{\text{vivo}}\mathcal{L}_{\text{vivo}}(\theta_{\text{vivo}}) + w_{\text{vitro}}\mathcal{L}_{\text{vitro}}(\theta_{\text{vitro}}) + \mathcal{L}_{\text{soft}}(\theta), \quad (44)$$

where θ_{vivo} and θ_{vitro} are the context-specific parameter vectors reconstructed from the joint optimizer vector and passed to the two likelihood components. For each soft-coupled natural-scale parameter Q , let

$$z_Q = T_Q(Q)$$

denote its transformed optimizer-scale value, where T_Q is the transformation used for that parameter during optimization. The separate *in vivo* and *in vitro* fits define context-specific transformed bounds $[l_Q^{\text{vivo}}, u_Q^{\text{vivo}}]$ and $[l_Q^{\text{vitro}}, u_Q^{\text{vitro}}]$. In the joint fit, these component-specific bounds are used only to define a single active joint admissible interval:

$$l_Q^{\text{joint}} = \min(l_Q^{\text{vivo}}, l_Q^{\text{vitro}}), \quad u_Q^{\text{joint}} = \max(u_Q^{\text{vivo}}, u_Q^{\text{vitro}}). \quad (45)$$

Thus, the original component-specific bounds record the provenance of the separate fits, whereas the joint optimizer uses the union interval $[l_Q^{\text{joint}}, u_Q^{\text{joint}}]$ as the active transformed-scale admissible range. During objective evaluation, the transformed values passed to the two components are reconstructed as

$$z_Q^{\text{vivo}} = c_Q + \frac{\delta_Q}{2}, \quad z_Q^{\text{vitro}} = c_Q - \frac{\delta_Q}{2}. \quad (46)$$

The reconstructed values must satisfy

$$z_Q^{\text{vivo}} \in [l_Q^{\text{joint}}, u_Q^{\text{joint}}], \quad z_Q^{\text{vitro}} \in [l_Q^{\text{joint}}, u_Q^{\text{joint}}]. \quad (47)$$

Parameter vectors that do not yield admissible reconstructed transformed values are outside the joint feasible domain and are not evaluated as valid likelihood points. The soft-coupling penalty is applied to the offset variable,

$$\mathcal{L}_{\text{soft}}(\theta) = \sum_{Q \in \mathcal{S}} \frac{c^2}{2} \left[1 - \exp \left\{ - \left(\frac{\delta_Q}{c\sigma_Q} \right)^2 \right\} \right], \quad (48)$$

where $\mathcal{S}$ is the set of soft-coupled parameters, σ_Q is the transformed-scale soft-coupling scale, and c controls the transition from near-quadratic behavior to saturation. For the reported fits, $\sigma_Q = 0.65$ for all 14 parameters and $c = 0.4$. For $|\delta_Q| \ll c\sigma_Q$, each term approaches $\frac{1}{2}(\delta_Q/\sigma_Q)^2$; for large separations it saturates at $c^2/2$, limiting the influence of a single strongly context-specific

parameter. Since

$$z_Q^{\text{vivo}} - z_Q^{\text{vitro}} = \delta_Q,$$

this penalty regularizes the transformed difference between the *in vivo* and *in vitro* parameter values. For log-transformed parameters, this corresponds to regularization of fold differences rather than absolute natural-scale differences.

Warm-start initialization for the joint fit. The joint analysis used six landscape-informed separate *in vivo* fits as warm-start representatives and paired each with the same best separate *in vitro* fit (seed 10) on the transformed optimizer scale. For each soft-coupled parameter Q , let $z_Q^{\text{vivo},0}$ and $z_Q^{\text{vitro},0}$ denote the transformed values from one such pair. The initial center and offset are set to

$$c_Q^0 = \frac{z_Q^{\text{vivo},0} + z_Q^{\text{vitro},0}}{2}, \quad \delta_Q^0 = z_Q^{\text{vivo},0} - z_Q^{\text{vitro},0}. \quad (49)$$

This initialization exactly reconstructs the two separate-fit transformed values:

$$z_Q^{\text{vivo},0} = c_Q^0 + \frac{\delta_Q^0}{2}, \quad z_Q^{\text{vitro},0} = c_Q^0 - \frac{\delta_Q^0}{2}.$$

Parameters that appear only in one component, such as context-specific observation-model parameters, are initialized from the corresponding separate-fit solution.

The warm start must be representable under the active joint union bounds in Eq. (45); otherwise, initialization fails explicitly. This ensures that the joint fit begins from the separate best-fit solutions only when those solutions are valid under the declared joint feasible domain.

Optimization and data inclusion. Optimization uses a global search followed by local refinement. Separate *in vivo* and *in vitro* searches each used 500 numerical seeds. In the joint analysis, each of the six warm-start pairs used 500 numerical seeds, for 3,000 optimizer runs in total. Seed-to-seed spread was used to assess numerical search robustness and was not interpreted as biological replication or inferential uncertainty. Retained *in vivo* observation times satisfy $0 < t_{i,1} < \dots < t_{i,n_i} \leq T_i^{(H)}$, so day-0 burden points are excluded and observations are right-censored at harvest. Only tumors with both burden and terminal chromosome-number data are retained. Terminal endpoint discrepancies are evaluated directly from the single-cell chromosome-number observations under the Gaussian-mixture likelihood in Eq. (35), and tumor-level endpoint losses are aggregated equally within each initial-ploidy cohort before applying cohort balancing as in Eq. (37).

Visualization and reporting For visualization/reporting in cell-number units, observed tumor volume (Eq. (27)), is converted to an approximate cell-count range using the specified diploid packing-density bounds:

$$N_{ij}^{\text{obs,low}} = V_{ij}^{\text{obs}} \rho_{2N,\text{min}}, \quad N_{ij}^{\text{obs,mid}} = V_{ij}^{\text{obs}} \sqrt{\rho_{2N,\text{min}} \rho_{2N,\text{max}}}, \quad N_{ij}^{\text{obs,high}} = V_{ij}^{\text{obs}} \rho_{2N,\text{max}}. \quad (50)$$

This conversion is used for plotting/interpretation only and is not part of the fitted likelihood, which is evaluated directly in the volume domain via Eq. (27).

1.4 Parameter Tables

The estimated calibration parameters are summarized in Table 1, and the fixed numerical quantities in the current model specification are summarized in Table 2. For the estimated parameters we report the model parameter in natural scale, the optimizer initialization value, the admissible calibration bounds, and the literature source that motivated the chosen range.

Whenever possible, parameter ranges were anchored to experimentally measured quantities reported in the literature. In particular, oxygen-related parameters (e.g. oxygen supply dynamics and hypoxia response) were constrained using values reported in experimental studies of tumor oxygenation and diffusion. Parameters governing chromosomal missegregation and whole-genome doubling were guided by estimates from tumor evolution studies.

Parameters for which no reliable biological measurements exist were assigned conservative engineering bounds to ensure numerical stability of the optimization while preserving sufficient flexibility for model fitting.

The estimated-parameter specification is expressed on the natural scale. For joint fits, shared *in vivo/in vitro* bounds are first merged as in Eq. (45); selected soft-coupled parameters are then represented by the center–delta construction in Eq. (46).

Supplementary Table 1: Estimated model parameters and calibration bounds for the current O_2 –glucose supply-demand model. Initial values and bounds are reported on the natural parameter scale. For Gaussian soft priors, the prior mean and prior standard deviation are reported on the calibration scale used in fitting; bounded uniform priors are shown as ‘–’. Hypoxia weighting uses $h(O_2) = O_{2,crit}^{n_O} / (O_{2,crit}^{n_O} + O_2^{n_O})$, where $O_{2,crit}$ is an independently estimated parameter.

Parameter	Initial value	Lower bound	Upper bound	Prior distribution	Prior mean	Prior SD	Unit	Description	Reference
λ_{max}	0.3512	0.2498	0.700	Bounded uniform	–	–	day ^{−1}	Maximum baseline division rate in the O_2 model.	Hafner et al. (2016); ²⁸ Bairoch (2018). ²⁹
$p_{mis,base}$	1.0×10^{-5}	1.0×10^{-6}	5.0×10^{-3}	Bounded uniform	–	–	probability	Baseline per-chromosome missegregation probability.	Thompson & Compton (2011); ³⁰ Gordon et al. (2012); ³¹ Ganem et al. (2009); ³² Ben-David & Amon (2020). ³³
p_{misseg}	1.50×10^{-2}	5.00×10^{-3}	0.400	Bounded uniform	–	–	probability	Stress-linked missegregation amplification scale in $p_{mis} = p_{mis,base} + p_{misseg} \mu_{eff} / (\mu_{eff} + k_{o,mis})$.	Gordon et al. (2012); ³¹ Brown & Wilson (2004); ³⁴ Bakhroum et al. (2018); ³⁵ Quinton et al. (2021). ³⁶
$k_{o,mis}$	3.0×10^{-3}	1.0×10^{-3}	2.0×10^{-2}	Bounded uniform	–	–	day ^{−1}	Half-saturation scale on μ_{eff} for death-linked missegregation amplification.	Phenomenological half-saturation scale for the fitted death-linked missegregation response; no direct quantitative literature bound is used.
s_{max}	0.800	0	1.000	Gaussian soft (truncated by bounds)	0.800	0.25	unitless	Maximum per-copy survival factor for symmetric buffering of missegregated daughters.	Fujiwara et al. (2005); ³⁷ Quinton et al. (2021); ³⁶ Santaguida & Amon (2015); ³⁸ Sheltzer & Amon (2011). ³⁹

Parameter	Initial value	Lower bound	Upper bound	Prior distribution	Prior mean	Prior SD	Unit	Description	Reference
β_{buf}	1.000	1.0×10^{-2}	10.000	Gaussian soft (truncated by bounds)	0.000	0.75	unitless	Ploidy-buffering strength for symmetric missegregation survival.	Fujiwara et al. (2005); ³⁷ Quinton et al. (2021); ³⁶ Santaguida & Amon (2015). ³⁸
n_{exp}	1.000	0.100	10.000	Gaussian soft (truncated by bounds)	0.000	0.75	unitless	Ploidy-buffering exponent for symmetric missegregation survival.	Fujiwara et al. (2005); ³⁷ Quinton et al. (2021); ³⁶ Sheltzer & Amon (2011). ³⁹
p_{WGD}	1.00×10^{-4}	1.00×10^{-5}	1.00×10^{-3}	Bounded uniform	—	—	probability	Constant per-division whole-genome-doubling probability.	Fujiwara et al. (2005); ³⁷ Bielski et al. (2018); ¹ Zack et al. (2013); ⁴⁰ Quinton et al. (2021). ³⁶
S_0	3.500	2.000	5.000	Gaussian soft (truncated by bounds)	0.544	0.35	% O ₂	Supply-demand oxygen baseline in the logarithmic O ₂ target relation.	Bertout et al. (2008); ⁴¹ McKeown (2014). ⁴²
κ_O	0.500	0.100	1.200	Gaussian soft (truncated by bounds)	-0.602	0.50	% O ₂	Oxygen-drop coefficient in the logarithmic O ₂ target model.	Bertout et al. (2008); ⁴¹ McKeown (2014). ⁴²
η	1.200	1.000	2.000	Gaussian soft (truncated by bounds)	0.0792	0.20	unitless	Exponent linking chromosome count to ploidy-weighted oxygen demand.	Phenomenological exponent estimated by the model to link ploidy and oxygen demand; no direct quantitative literature bound is used.
ρ_{2N}	56000	50000	500000	Gaussian soft (truncated by bounds)	4.627	0.15	cells mm ⁻³	Diploid-cell packing density used when converting cell counts into tumor volume.	Waclaw et al. (2015). ⁴³
α_{O_2}	1.500	0.500	4.000	Bounded uniform	—	—	unitless	Strength of the soft oxygen-mediated high-ploidy growth penalty.	McKeown (2014); ⁴² Brown & Wilson (2004). ³⁴
γ_{growth}	3.000	1.500	4.500	Bounded uniform	—	—	unitless	Exponent shaping how strongly the proliferation penalty increases with chromosome count.	Sheltzer & Amon (2011). ³⁹
μ_{HP}	1.5×10^{-3}	1.0×10^{-5}	8.0×10^{-3}	Gaussian soft (truncated by bounds)	-2.824	0.35	day ⁻¹	Soft thresholded high-ploidy hypoxia death strength.	Bertout et al. (2008); ⁴¹ Brown & Wilson (2004); ³⁴ McKeown (2014). ⁴²
γ_{μ}	2.000	1.200	3.500	Gaussian soft (truncated by bounds)	2.000	0.50	unitless	Exponent controlling how sharply hypoxia-linked death rises above the diploid reference state.	Phenomenological exponent controlling hypoxia-linked death steepness; no direct quantitative literature bound is used.
$O_{2,\text{crit}}$	1.000	0.100	2.500	Bounded uniform	—	—	% O ₂	Hill critical oxygen scale for oxygen-response hypoxia weighting.	Bertout et al. (2008); ⁴¹ McKeown (2014). ⁴²
n_O	1.000	0.500	5.000	Bounded uniform	—	—	unitless	Hill exponent controlling the steepness of oxygen-response hypoxia weighting around $O_{2,\text{crit}}$.	Hill-shape exponent for oxygen-response steepness; empirical response-shape parameter rather than directly measured quantity.

Parameter	Initial value	Lower bound	Upper bound	Prior distribution	Prior mean	Prior SD	Unit	Description	Reference
k_{clear}	1.0×10^{-4}	1.0×10^{-5}	0.100	Gaussian soft (truncated by bounds)	-4.000	0.35	day ⁻¹	Clearance rate for mass already routed into the dead-cell compartment.	Phenomenological dead-cell-compartment clearance rate; empirical model parameter without direct measured bound.
σ_{burden}	0.200	0.100	0.600	Bounded uniform	–	–	unitless	Standard deviation of the log-normal burden observation model.	Eisenhauer et al. (2009). ⁴⁴

Supplementary Table 2: Fixed numerical settings used by the current O₂–glucose supply-demand model and calibration workflow. Values are reported on the natural scale.

Parameter	Value	Unit	Description	Reference
τ_{O_2}	0.1	day	Oxygen relaxation timescale.	Fixed numerical relaxation setting; empirical workflow choice rather than a literature-derived parameter.
$O_{2,\text{cap}}$	5.0	% O ₂	Maximum oxygen level used to cap the supply-demand oxygen trajectory, inherited from the S_0 upper bound.	Bertout et al. (2008); ⁴¹
$O_{2,\text{min}}$	0	% O ₂	Lower oxygen floor used in the logarithmic oxygen target relation.	McKeown (2014). ⁴² Physical/numerical anoxia floor used to bound oxygen values; model constraint rather than literature-estimated threshold.
N_{ref}	10^6	cells	Reference viable burden scale used in the oxygen supply-demand relation.	Oxygen-feedback normalization constant; empirical model scaling value rather than a measured biological quantity.
N_{unit}	22	chromosomes	Chromosome count corresponding to ploidy 1.	Model convention defining one haploid autosome set as ploidy 1; definition rather than literature-estimated parameter.
N_{min}	22	chromosomes	Minimum chromosome-count state retained in the discrete state space.	Discrete state-space lower bound anchored to the model ploidy convention; grid-design choice.
N_{max}	154	chromosomes	Maximum chromosome-count state retained in the discrete state space.	Bielski et al. (2018); ¹ Zack et al. (2013). ⁴⁰
Δt	0.05	day	Simulation time step.	Simulation time step selected for numerical resolution and stability; computational setting rather than biological parameter.
K	10^{12}	cells	Carrying capacity in the logistic crowding term.	Waclaw et al. (2015). ⁴³
$N(0)$	10^6	cells	Initial total tumor population size.	Initial-condition scale for the simulation/calibration workflow; empirical setup choice.
σ_{ploidy}	0.12	unitless	Fixed endpoint chromosome-number observation noise in the harvest likelihood.	Fixed likelihood observation-noise term; empirical measurement-error setting in the calibration workflow.
$\varepsilon_{\log B}$	10^{-12}	unitless	Numerical offset added before the burden log transform.	Numerical log-offset to prevent log(0); computational safeguard.
$N_{\text{min,pop}}$	10^{-12}	unitless	Minimum population floor used for numerical stability.	Population floor used to avoid numerical underflow or zero states; computational safeguard.

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